ANIEZE
PARIS

FOLD-PAINT SPLIT SET

REVISION B · 11 NOV 2026 · REPLACES ALL EARLIER COPIES

Rev B · 11 Nov 2026 · Added hem circumference: CC on the overshirt (= 2 × C, pleats closed), LOC on the trouser (= 2 × LO); no measurement changed. ANIEZE logo on every page.

Rev A · 11 Nov 2026 · First issue (colourway 1, all wool).



colourway 1 · Indigo / Stone · Rust + Caramel

Two everyday wool pieces joined by ONE hand-painted composition. A boxy indigo overshirt whose lower half is a separate panel of deep knife pleats; a wide, high-rise stone trouser with deep pleats that pools on the floor. Rust and caramel paint, feathered and dye-based, starts on the shirt's pleats and runs on across the trouser's front pleats and back thighs, fading out by the knee. Each piece stays ≈ 80 % its own base colour; every set is unique.

Styles	P07-FP01 overshirt · P07-FP02 trouser · Daily Drop D1111-1 · cw1
Sizes	Unisex XS–2XL, base M (chest 108.1 / waist 101.8 / hip 106.9, 174 cm)
FP01	deep indigo wool gabardine 220–240 g/m² · 36 knife pleats · 8 snaps · buy ≈ 3.7 m (M)
FP02	warm stone wool gabardine 280–320 g/m² · 2 pleats per leg, permanent set · half-lined front to the knee · buy ≈ 2.8 m (M)
Paint	FP01: DISCHARGE painting (illuminating acid dyes replace the indigo) · FP02: direct acid-dye painting · steam-fixed 102 °C · both pieces painted together on one template
Hardware	Metal, nickel-free: snaps, #5 zip, hook + bar, inside button · A silver-tone (confirm) / B gunmetal
Weight	FP01 ≈ 0.79 kg · FP02 ≈ 0.83 kg at M (estimates)
Production	Sialkot, Pakistan · imported cloth · proto panels first

P0 spec for the first proto panels and toile. Measurements are designer estimates until the fitting; the painting route is to be validated on proto panels.



The design image shows the look; the drawings and tables rule. Pleat seam: the design image places it at the upper abdomen, ≈ 46 % of the shirt length → PS = 33 cm below HPS at M (≈ 10 cm above the natural waist).











Flat lay: the small dye splatter dots are an artefact of the design image: the spec is soft feathered edges only, no splatter. Snaps design image bright silver-tone: finish A. Details: painted pleats, pleat seam + snap, trouser pleat with dye fade, waistband / hook + bar / slant pocket.

Colour	Role	Target description	Nearest Pantone TCX	Alternative
Deep indigo	FP01 base (wool gabardine 220–240 g/m²)	Deep indigo / dark navy, slight red-violet cast, mid-dark (target L* ≈ 22–26); dyed with DISCHARGEABLE acid dyes	**19-3953 TCX Sodalite Blue** *(nearest)*	19-4027 TCX Estate Blue
Warm stone	FP02 base (wool gabardine 280–320 g/m²)	Warm light stone / greige-beige with a yellow cast (target L* ≈ 72–76), even, no grey-green	**14-1112 TCX Pebble** *(nearest)*	15-1116 TCX Safari
Rust	paint, cores of each shape	Deep warm red-orange brown like oxidised iron; the darker of the two paint colours	**18-1248 TCX Rust** *(nearest)*	18-1340 TCX Potter's Clay
Caramel	paint, halos and feathered edges	Golden tan-orange, lighter and yellower than rust; feathers out into the base	**16-1439 TCX Caramel** *(nearest)*	16-1342 TCX Buckskin

All Pantone references are NEAREST matches only: confirm with lab dips. FP01 colours come from DISCHARGE paste with illuminating acid dyes on the indigo; FP02 colours from direct acid-dye paint on the stone. Both must match the same rust and caramel standards within grey scale 4 (ISO 105-A02) under D65, approved side by side in one lab-dip round.

Other colourways (same pattern + paint map, recipes re-developed per colourway)


cw2 · Black / Deep green · Verdigris + Stone

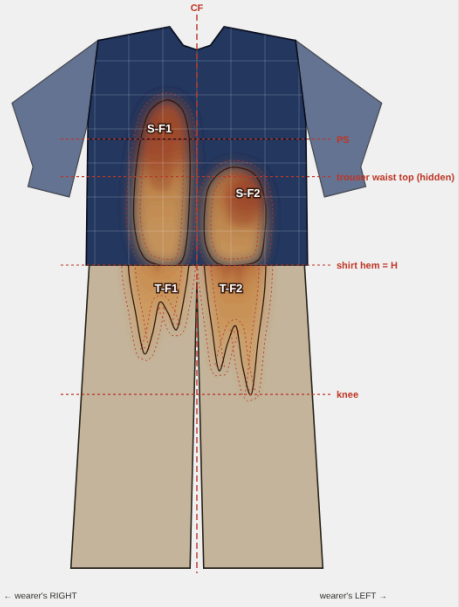

cw3 · Amethyst / Charcoal · Russet + Caramel


cw4 · Caramel / Chocolate · Indigo + Charcoal


cw5 · Ecru / Indigo · Amethyst + Charcoal

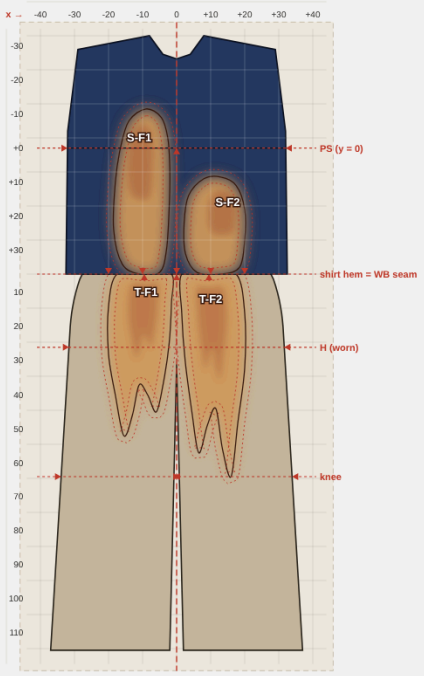
AS WORN · FRONT

overshirt over the trouser · trouser waist 44 cm below HPS · shirt hem = H



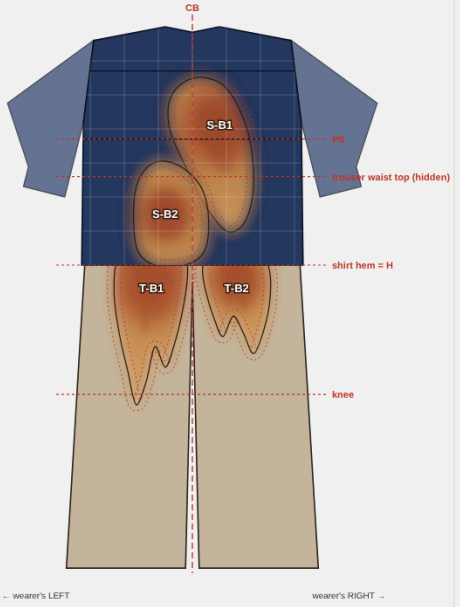
LAID FLAT ON THE TEMPLATE · FRONT

front paint units: shirt (x from CF, y from PS) · trouser legs (y from the waistband top)



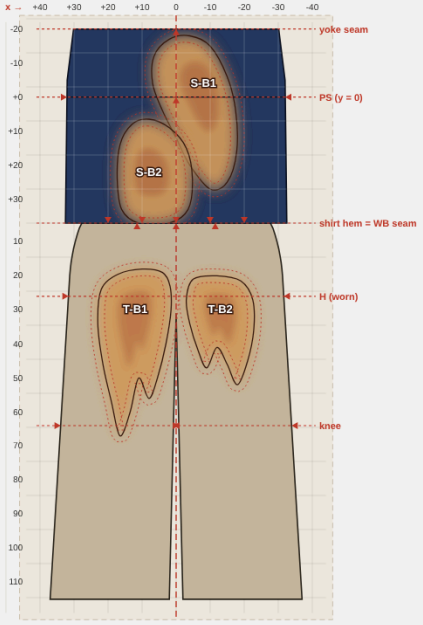
AS WORN · BACK

seen from behind: the wearer's RIGHT is on the image right



LAID FLAT ON THE TEMPLATE · BACK

back paint units: back body below the yoke + pleats · back legs



shape outline at M (curve through the template points, table P-1) · ±2 cm envelope: every piece unique inside it · registration notch (3 mm, in the allowance) = template pin point · CF / CB · grid 10 cm · x + = wearer's LEFT

Rust cores, caramel halos, edges feathered 3–6 cm, NO splatter. Pleats basted shut while painting (faces painted, underlays stay base colour). Plackets, yoke, sleeves, collar, waistband, loops: never painted.

Template: shirt unit above, trouser legs below, the shirt hem touching the trouser waistband seam (laid flat). As worn the shirt hem falls at H = 26 cm below the trouser waistband top: the streaks run on across it. Other sizes: scale per table P-2.

Shape	Where	y ref	Outline points at M: (x cm from CF/CB, + = wearer's LEFT ; y cm below the reference) · tolerance ±2 cm	Rust core	cm²
S-F1	Overshirt front · wearer's RIGHT of CF · from ≈ 11 cm above the pleat seam (spills onto the upper front) down to the hem	PS	(-9, -11.5) · (-4.5, -9) · (-2.6, -3) · (-2, 6) · (-2.2, 18) · (-3, 30) · (-5, 36.5) · (-11, 37) · (-16, 34) · (-18.5, 24) · (-18, 10) · (-16.5, 0) · (-14, -8)	(-11, -3)	683
S-F2	Overshirt front · wearer's LEFT of CF · from ≈ 9 cm below the pleat seam to the hem	PS	(8, 9) · (13, 8.5) · (17.5, 11) · (20, 18) · (20, 27) · (18.5, 35) · (13, 37) · (6, 36.5) · (2.6, 31) · (2.2, 22) · (3.5, 14)	(15, 16)	450
S-B1	Overshirt back · upper lobe · from 2 cm below the yoke seam across the wearer's RIGHT back, onto the pleats	PS	(-4, -18) · (2, -17) · (6.5, -12) · (6.9, -4) · (4, 4) · (0, 12) · (-4, 20) · (-10, 27) · (-15, 25) · (-17.8, 16) · (-17.5, 4) · (-15, -6) · (-10, -15)	(-8, -4)	779
S-B2	Overshirt back · lower lobe (joins the upper lobe) · wearer's LEFT of CB, on the pleats down to the hem	PS	(8, 6.5) · (13, 8) · (16.5, 14) · (17.3, 24) · (16, 33) · (11, 37) · (2, 37) · (-3.5, 33) · (-4.8, 24) · (-3, 15) · (2, 9)	(8, 22)	571
T-F1	Trouser front · wearer's RIGHT leg · over both pleats from the waistband seam, fading out at mid-thigh (≈ 52 cm)	WB top	(-1.5, 4.5) · (-9, 4.5) · (-17.5, 4.5) · (-20, 14) · (-20, 28) · (-18, 40) · (-15.5, 52) · (-13, 46) · (-11, 37) · (-8.5, 40) · (-6, 45) · (-4, 37) · (-2.2, 25) · (-1.5, 13)	(-9, 10)	671
T-F2	Trouser front · wearer's LEFT leg · over both pleats from the waistband seam, fading out at the knee (≈ 64 cm)	WB top	(1.5, 4.5) · (9, 4.5) · (17.5, 4.5) · (20, 16) · (20, 32) · (18, 48) · (16, 64) · (13.5, 56) · (11.5, 44) · (9, 49) · (6.5, 57) · (4.5, 45) · (2.5, 30) · (1.5, 16)	(10, 12)	823
T-B1	Trouser back · wearer's LEFT thigh · from ≈ 5 cm below the back pocket, fading out just above the knee (≈ 67 cm)	WB top	(4, 19) · (10, 18) · (17, 19.5) · (22, 24) · (23, 34) · (21.5, 46) · (19, 57) · (16.5, 67) · (13.5, 60) · (11, 50) · (8, 56) · (5.5, 50) · (3, 40) · (1.5, 30) · (1.8, 23)	(12, 30)	715
T-B2	Trouser back · wearer's RIGHT thigh · from ≈ 7 cm below the back pocket, fading out at mid-thigh (≈ 52 cm)	WB top	(-5, 21) · (-12, 20) · (-19, 21.5) · (-22.5, 27) · (-22.8, 36) · (-21, 45) · (-18, 52) · (-15, 46) · (-12, 41) · (-9, 47) · (-6.5, 42) · (-4, 34) · (-3, 27)	(-13, 30)	452

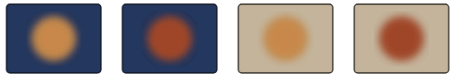
Size	shirt x	shirt y ↓PS	shirt y ↑PS	trouser x	trouser y
XS	0.938	0.989	0.952	0.923	0.971
S	0.969	0.995	0.976	0.962	0.986
M	1.000	1.000	1.000	1.000	1.000
L	1.031	1.005	1.024	1.038	1.014
XL	1.062	1.011	1.048	1.077	1.029
2XL	1.109	1.024	1.079	1.135	1.047

1. Each garment stays ≈ 80 % base colour: paint covers ≤ 25 % of the visible garment (QA checks it from these outlines).
2. Rust sits in the core of each shape (upper part), caramel around it; rust : caramel ≈ 40 : 60 by area. Edges feather out over 3–6 cm into the base. NO splatter, no drips, no hard edges (the dots in the flat-lay design image are an artefact).
3. Tolerance: every outline ±2 cm (the template shows the inner and outer envelope). Inside the envelope every piece is unique: brush marks, feathering and the rust / caramel balance vary by hand.
4. Paint is applied with the pleats BASTED SHUT: only the pleat faces take colour, the underlays keep the base colour, so indigo / stone flashes between painted faces when the pleats move (the 'fold' of fold-paint).
5. Plain (never painted): overshirt plackets, collar, stand, yoke, sleeves, cuffs; trouser waistband, belt loops, fly shield, pocket facings and bags. Paint stays ≥ 1.5 cm inside the slant pocket edge, ≥ 1 cm from the trouser CF and ≥ 5 cm below the back pocket jets.
6. Other sizes: one template per size, scaled from M (table below). Overshirt: x × B/B(M), y below the pleat seam × PL/PL(M), y above it × PS/PS(M). Trouser: x × W/W(M), y × (OS − WBH)/(OS − WBH)(M).

HOW THE PAINTING IS DONE · route A (to be validated on proto panels)

FP01 indigo overshirt: DISCHARGE painting (illuminating rust + caramel replace the indigo) · FP02 stone trouser: direct acid-dye painting · both pieces painted together on one template, then steam-fixed

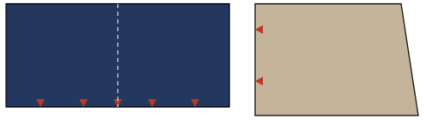
1 LAB DIPS + STRIKE-OFFS



→ ISO 105-X12 · D01 · B02 · strike-off per fabric

Approve the four colours (indigo + stone bases, rust + caramel on each) and one full-route strike-off per fabric. Tests: ISO 105-X12, D01, B02. Nothing is cut before approval.

2 CUT + NOTCH



3 mm notches in the allowance · thread-trace CF / PS

Cut each size's pieces, snip the 3 mm registration notches in the allowances, thread-trace CF / CB and the pleat seam line. Keep the pieces of one set together (set number).

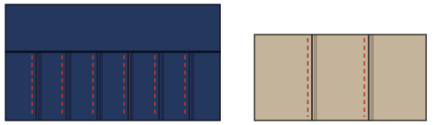
3 PLEAT + PERMANENT SET



fold → chemical set (green, MEA-sulphite type) → press

At the pleating / finishing house: form the knife pleats (overshirt panel: 36 pleats, face ≈ 3 cm, depth 6 cm; trouser: 2 per leg + crease), apply the chemical set (MEA-sulphite type, Si-Ro-Set principle), press. To be validated on proto panels.

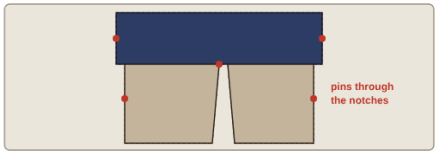
4 BASTE SHUT + BUILD PAINT UNITS



basting 1 cm from every fold · shirt unit joined at PS

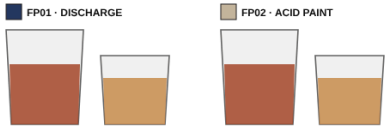
Baste every pleat shut with polyester thread 1 cm from the fold, full length. Join the overshirt units at the pleat seam: upper front + pleated front, back body + pleated back (yoke and plackets NOT on yet). Trouser: front and back legs as single panels.

5 LAY UP ON THE TEMPLATE



Padded table covered in plastic; the 1:1 template for the size; pin the units through the notches: overshirt above, trouser below, shirt hem touching the trouser waistband line (laid-flat layout). Plastic sheet under every unit.

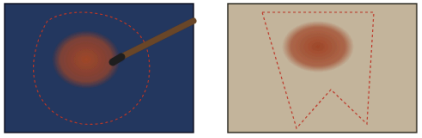
6 MIX THE PASTES



all thickened with sodium alginate · pH 4–5

FP01 indigo: DISCHARGE paste = non-alkaline discharge agent for wool (Decrolin-type) + illuminating acid dyes (rust, caramel) in sodium alginate. FP02 stone: milling / 1:2 metal-complex acid dyes in sodium alginate, citric acid to pH 4–5. Thin versions for the spray.

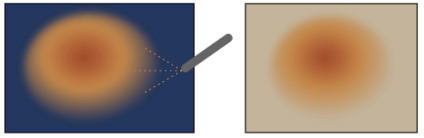
7 PAINT: RUST CORES



inside the inner envelope · one gesture across both pieces

Brush the rust cores first, inside the inner envelope, following the template outline at ±2 cm. Work across the shirt hem onto the trouser in one gesture so the shapes run on.

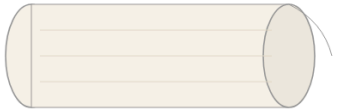
8 PAINT: CARAMEL + FEATHER



feather 3–6 cm · NO splatter · ≈ 80 % base colour

Brush caramel around the cores, then feather every edge over 3–6 cm with a damp brush and a fine spray. No splatter, no drips. Check ≈ 80 % base colour per piece. Every piece is unique inside the envelope.

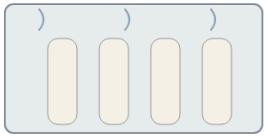
9 DRY + ROLL



clean unprinted paper · painted faces never touch

Let the paste dry flat (no contact between units). Roll each unit in clean unprinted paper so painted faces never touch.

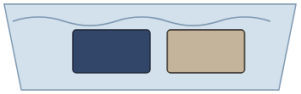
10 STEAM-FIX



saturated steam 102 °C · 30–45 min · no drips

Saturated steam at 102 °C for 30–45 min (discharge: per the dye-house recipe), rolls standing, no condensate drips. Cool slowly.

11 RINSE + NEUTRALISE



cold → ≤ 40 °C non-ionic soap · NO agitation (felting)

Cold rinse, then ≤ 40 °C with a non-ionic soap, NO agitation (wool felts); the discharged FP01 units get an extra rinse to remove reducing-agent residues and smell. Dry flat.

12 RELEASE + RE-SET + QC



pull basting · re-press + re-set · check envelope + colour · tag

Pull the basting, re-press and re-set the pleats and creases. Check each piece against the template envelope and the colour standard, tag the set number, photograph it, then make the garments.

Fallback B (only if route A fails on the proto panels): paint the panels FLAT, steam-fix, rinse, then pleat and set. The composition then splits across pleat faces and underlays (the shapes look broken into stripes when the pleats close) and registration across the pleats is lost. Not the design intent.

Every outline: ±2 cm from the template (paint map). Shapes, colours and tests: see the paint map, the colour standards and the test plan. Not pigment: dyes only, the hand stays soft.

Registration marks (notches) + template

Overshirt front unit	3 mm notches in the hem allowance at CF and at x = ±10 / ±20 cm; notch at CF and at each side seam on the pleat seam (PS); thread-trace CF
Overshirt back unit	notches in the hem allowance at CB and x = ±10 / ±20 cm; CB notch on the pleat seam and on the yoke edge; thread-trace CB
Trouser front legs	notch at CF and on the pleat-1 fold (crease line) at the waistband seam; notches on the side seam and inseam at H (shirt hem as worn, 26 cm below the waistband top at M) and at the knee (64 cm)
Trouser back legs	notch at CB and at the back pocket centre on the waistband seam; side seam + inseam notches at H and at the knee
Template	1:1 sheet per size (Mylar or heavy kraft) on the padded table: outlines of every shape with the ±2 cm envelope dashed, CF / CB lines, pleat seam, H and knee lines, and pin points that match every notch. Pieces are pinned through the notches only (allowance), never in the visible field.

Fallback

Fallback B (only if route A fails on the proto panels): paint the panels FLAT, steam-fix, rinse, then pleat and set. The composition then splits across pleat faces and underlays (the shapes look broken into stripes when the pleats close) and registration across the pleats is lost. Not the design intent.

Colour + fastness tests (before cutting bulk)

Lab dips	Rust + caramel on BOTH fabrics: discharge (illuminating dyes) on the indigo, direct acid paint on the stone; plus the indigo and stone bases. Approve all four against the colour standards, side by side, D65.
Strike-off	A 50 × 50 cm painted, pleated, steamed and rinsed panel of each fabric through the FULL route (pleat set → paint → steam → rinse → re-set), checked for shade, feathering, pleat sharpness and hand.
Colour fastness to rubbing	ISO 105-X12 on the painted zones of both fabrics: dry ≥ 4, wet ≥ 3 (grey scale for staining).
Colour fastness to dry-cleaning	ISO 105-D01 (perchloroethylene): change ≥ 4, staining ≥ 4; also check the pleat set and the snap finish after one cycle.
Colour fastness to light	ISO 105-B02 (xenon arc): ≥ 4 (blue wool scale) for rust, caramel, indigo and stone.
Steam test on fused parts	Fused collar, cuff, placket and waistband samples through 45 min saturated steam and one dry-clean cycle: no bubbling or delamination.

Production route (desk research, see sourcing)

Cloth
No Pakistani worsted-gabardine weaver confirmed → import the wool to Sialkot: Nona Source or Zerow for small runs (≈ 50 sets); Marzotto, VBC or Yünsa for ≈ 300 sets. Indigo must be commissioned in DISCHARGEABLE acid dyes (PFD / ecru base).

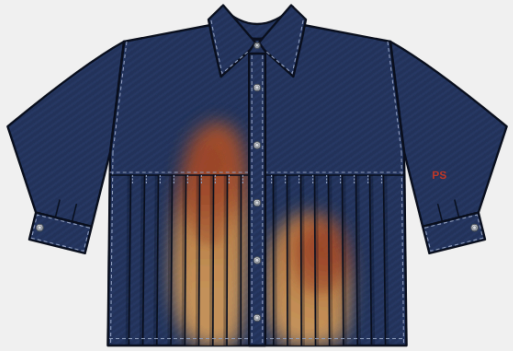
Pleating + permanent set
No Pakistani pleating house confirmed → Omniapiega (Italy), Maison du Pli or Atelier Lognon (Paris), or the factory installs a pleating press + steam cabinet with a chemical-set supplier.

Paint + steam-fix
No painted-garment steam house confirmed in Pakistan → the factory sets up a painting table + a star / cottage steamer (102 °C saturated steam), trained on the strike-offs.

Testing
SGS Karachi, Bureau Veritas Lahore or TTI Lahore: ISO 105-X12, D01, B02 (+ EN 1811 / REACH on the hardware).

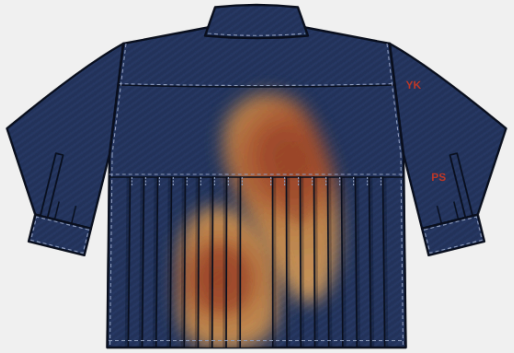
FRONT

indigo wool · 6 front snaps (stand + 5) · pleat seam PS · 9 knife pleats each side, folds toward the side seams



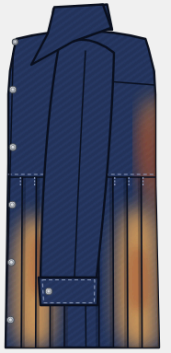
BACK

yoke · 9 + 9 knife pleats turning away from CB · paint from below the yoke onto the pleats



LEFT SIDE

dropped shoulder · snap cuff · pleats turn toward the side seam front and back



Indigo = wool gabardine · rust / caramel = hand-painted (discharge) composition, feathered edges, see the paint map · dashes = 6 mm topstitch · short dashes beside each fold = 2 cm edge-stitch · silver = nickel-free snaps

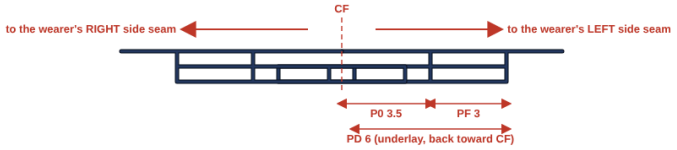
Knife pleats all round the lower half: 9 per front half and 9 per back half = 36 pleats. Face PF ≈ 3 cm, depth PD 6 cm (intake 12 cm per pleat, 432 cm in all at M, so the panel is cut ≈ 560 cm round before pleating). Every fold turns TOWARD THE NEAREST SIDE SEAM: on the wearer's LEFT of CF the folds face the wearer's left side seam, on the wearer's RIGHT of CF they face the right side seam; the same on the back, so CB reads as a flat 7 cm panel between two pleats turning away from it (an inverted box look). First fold P0 = 3.5 cm from CF / CB, last fold ≈ 1.5 cm before the side seam. Each pleat is edge-stitched shut (1.5 mm from the fold) for the first PE = 2 cm below the pleat seam, then hangs free. Because PD > PF the underlays overlap: ≈ 5.0 layers sit under one point of the pleat seam at M, so the seam allowance is graded and Hong Kong bound.

A · KNIFE PLEATS + EDGE-STITCH

36 pleats, folds toward the side seams both sides of CF / CB · face 3 · depth 6 · edge-stitched 2 cm

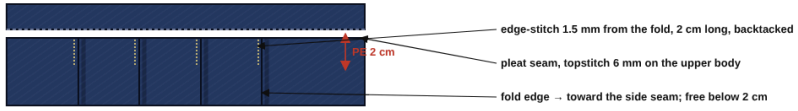
indigo wool gabardine 230 g/m² · woven fusible · cupro bias (Hong Kong) · stitching

① section through the pleat seam level (seen from below, schematic: layers spread apart)



Fold edges point to the nearest side seam on both sides of CF (and of CB). Face ≈ 3 cm, depth 6 cm → intake 12 cm per pleat, 432 cm in all;
the underlays overlap (PD > PF): ≈ 5 layers under the seam. Panel cut ≈ 560 cm round (4 quarter pieces, joins hidden in a pleat inner fold / in the side seam).

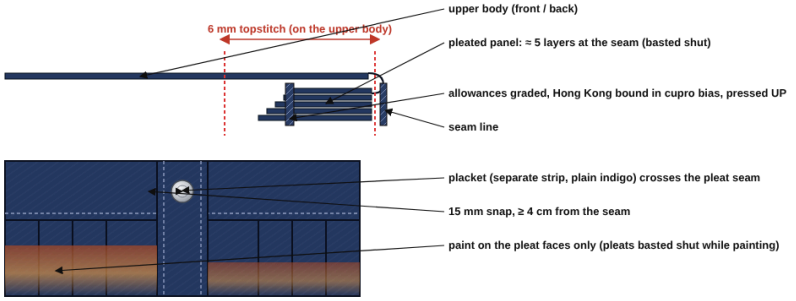
② plan at the pleat seam (wearer's left front, 4 pleats) · edge-stitch 2 cm, then free



B · PLEAT SEAM

Level all round at PS ≈ 33 cm below HPS · Hong Kong bound · topstitched 6 mm · placket crosses it

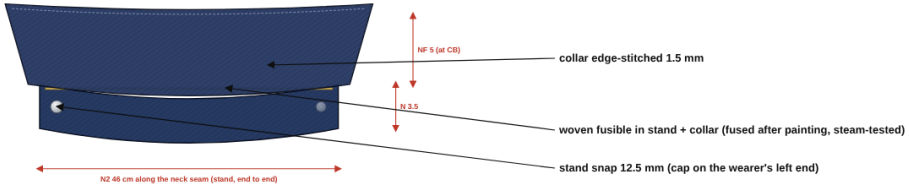
indigo wool gabardine 230 g/m² · woven fusible · cupro bias (Hong Kong) · stitching



C · COLLAR + STAND

Point collar on a stand · interfaced · stand snap

indigo wool gabardine 230 g/m² · woven fusible · cupro bias (Hong Kong) · stitching

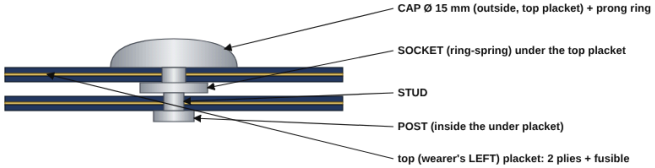


Point collar on a stand: collar point NP 7.5 cm, stand 3.5 cm at CB. Upper collar + under collar, both interfaced; stand edge-stitched 1.5 mm, collar 1.5 mm. Never painted.

D · SNAP SET (ring-spring, nickel-free)

Cap + socket on the top placket · stud + post on the under placket

indigo wool gabardine 230 g/m² · woven fusible · cupro bias (Hong Kong) · stitching



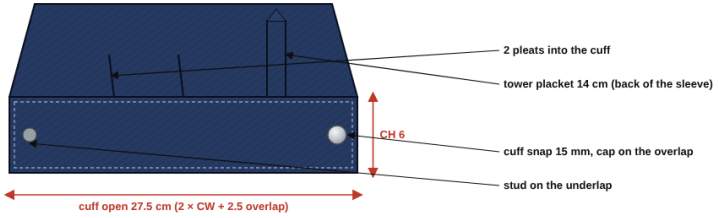


8 sets: stand 1 (12.5 mm), placket 5 (15 mm) at S1 = 14 cm below HPS then every SN = 12.5 cm, cuffs 2 (15 mm). Set AFTER painting, steaming and pressing, through 3–4 plies of wool + fusible; post length to suit the stack (2–4 mm). Fusible patch behind every snap.
Nickel-free; open / close force 15–30 N; pull ≥ 90 N.

E · CUFF + SLEEVE PLACKET

Folded 6 cm cuff · 1 snap · tower placket · 2 pleats

indigo wool gabardine 230 g/m² · woven fusible · cupro bias (Hong Kong) · stitching



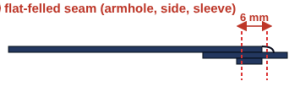
Folded cuff (outer + inner in one), woven fusible on the outer half, edge-stitched 1.5 mm all round. CW = 12.5 cm closed (cuff − wrist ≈ 8 cm, an overshirt cuff).

F · SEAMS, YOKE, HEM

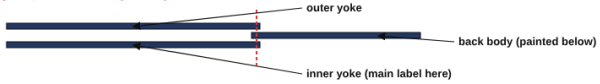
Flat-felled · yoke burrito · hem double-turned before pleating

indigo wool gabardine 230 g/m² · woven fusible · cupro bias (Hong Kong) · stitching

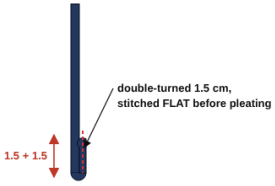
① flat-felled seam (armhole, side, sleeve)



② yoke, outer + inner (burrito)

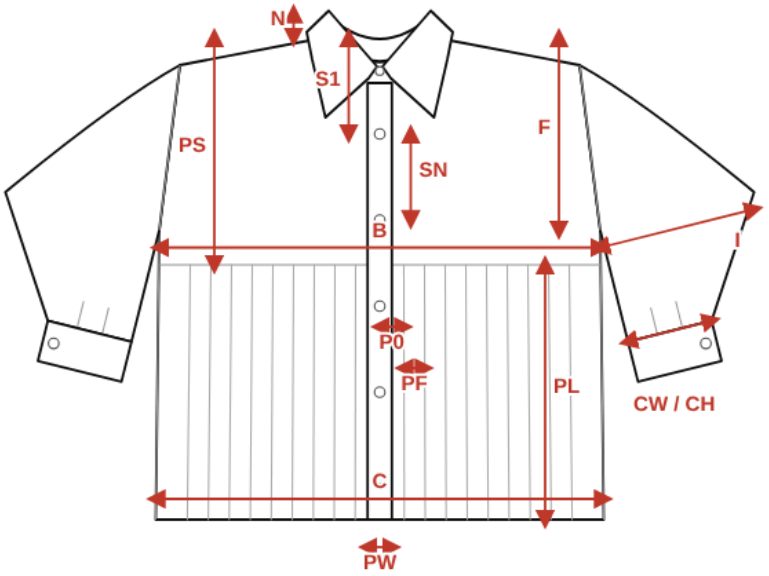


③ pleated-panel hem

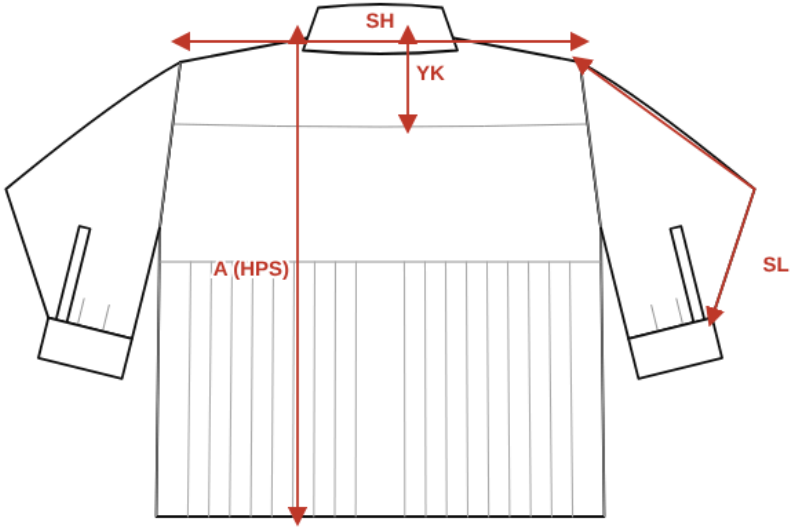


Unlined garment: flat-felled seams (clean inside), pleat seam + pleated-panel joins Hong Kong bound. Hem folds crisply into every pleat after the set.

FRONT · points of measure (M)



BACK



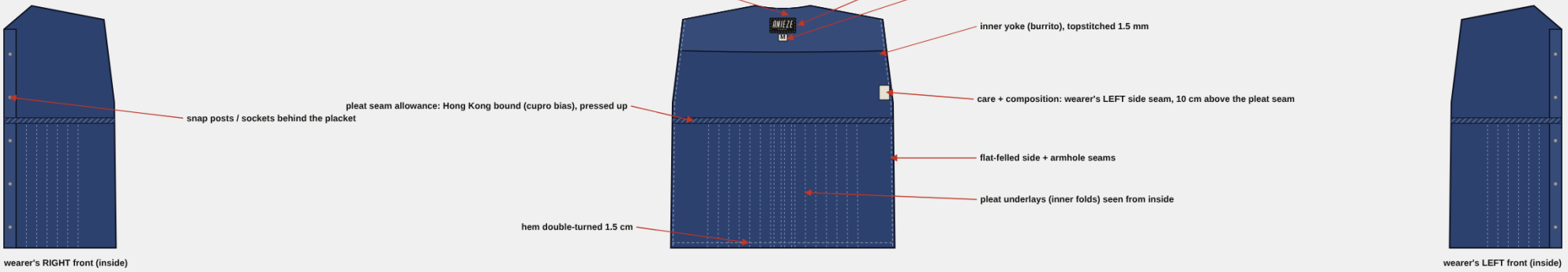
Snaps (M)	Size	Qty	Note
Collar stand (closes the neck)	12.5 mm cap	1	cap + socket on the wearer's LEFT stand end, stud + post on the right
Front placket, wearer's LEFT over right	15 mm cap	5	S1 below HPS, then every SN; cap + socket on the top (left) placket, stud + post on the under (right) placket
Cuff (one per sleeve)	15 mm cap	2	cap on the cuff overlap, stud on the underlap
Total 8	placket snaps at 14 · 26.5 · 39 · 51.5 · 64 cm below HPS · pleat folds at ±(P0 + k·PF) = ±3.5, ±6.5 … ±27.5 cm from CF / CB		

Code	Point of measure	How to measure	Tol	Grade	XL→2XL	XS	S	M	L	XL	2XL
A	Back length from HPS	High point shoulder (neck seam at the shoulder line) straight down to the hem edge at CB, laid flat, pleats closed	±1	1	1.5	68	69	70	71	72	73.5
B	1/2 chest	Laid flat, snaps closed, side to side 2.5 cm below the underarm point	±1.5	2	3	60	62	64	66	68	71
C	1/2 hem sweep, pleats closed	Laid flat, snaps closed, pleats lying shut, straight across the hem edge	±2	2	3	61	63	65	67	69	72
CC	Hem circumference, pleats closed	Snaps closed, pleats lying shut, all the way round the hem edge (= 2 × C)	±3	4	6	122	126	130	134	138	144
SH	Shoulder width (dropped)	Shoulder point to shoulder point (armhole seam at the shoulder line), straight across the back	±1	1.5	2	55	56.5	58	59.5	61	63
F	Armhole depth from HPS	HPS straight down to the underarm point (armhole seam meets side seam)	±1	1	1.2	26	27	28	29	30	31.2
SL	Sleeve length from shoulder point	Shoulder point along the top of the sleeve to the cuff edge, cuff included	±1	1	1	55	56	57	58	59	60
I	1/2 bicep	Laid flat, 2.5 cm below the underarm point, perpendicular to the sleeve centre	±1	0.8	1	21.4	22.2	23	23.8	24.6	25.6
CW	Cuff, closed, laid flat	Edge to edge across the closed cuff (snap done up) = 1/2 cuff circumference	±0.5	0.4	0.5	11.7	12.1	12.5	12.9	13.3	13.8
CH	Cuff height	Cuff seam to the cuff edge	±0.3	0	0	6	6	6	6	6	6
N2	Collar stand length at neck seam	Along the neck seam of the stand, end to end (incl. the placket extension)	±1	1.2	1.5	43.6	44.8	46	47.2	48.4	49.9
N	Collar stand height at CB	Neck seam to the stand/collar seam at CB	±0.2	0	0	3.5	3.5	3.5	3.5	3.5	3.5
NF	Collar height at CB	Stand/collar seam to the collar edge at CB	±0.3	0	0	5	5	5	5	5	5
NP	Collar point length	Stand/collar seam to the collar point, along the collar end edge	±0.5	0	0.3	7.5	7.5	7.5	7.5	7.5	7.8
PW	Front placket width	Edge to edge across the finished placket (centred on CF)	±0.2	0	0	3.5	3.5	3.5	3.5	3.5	3.5
S1	First placket snap below HPS	HPS straight down to the centre of placket snap 1 (the stand snap is extra, centred on the stand)	±0.5	0.3	0.3	13.4	13.7	14	14.3	14.6	14.9
SN	Placket snap spacing	Centre to centre, 5 placket snaps	±0.5	0.25	0.3	12	12.25	12.5	12.75	13	13.3
YK	Back yoke depth at CB	HPS level straight down CB to the yoke seam	±0.5	0.4	0.5	12.2	12.6	13	13.4	13.8	14.3
PS	Pleat seam from HPS	HPS straight down to the horizontal pleat seam (level front, side and back)	±1	0.8	1	31.4	32.2	33	33.8	34.6	35.6
PL	Pleated panel length	Pleat seam straight down to the hem edge at CF and CB (A = PS + PL)	±1	0.2	0.5	36.6	36.8	37	37.2	37.4	37.9
P0	First pleat fold from CF / CB	Along the pleat seam, CF (CB) to the fold of the first knife pleat, each side	±0.3	0	0	3.5	3.5	3.5	3.5	3.5	3.5
PF	Pleat face	Along the pleat seam, fold to fold of neighbouring pleats (9 pleats per front half and per back half)	±0.2	0.11	0.17	2.78	2.89	3	3.11	3.22	3.39
PD	Pleat depth	Fold to the inner (under) fold of one pleat, at the pleat seam; intake per pleat = 2 x PD	±0.3	0	0	6	6	6	6	6	6
PE	Pleat edge-stitch length	Pleat seam down to the end of the edge-stitch that holds each pleat shut	±0.2	0	0	2	2	2	2	2	2

Code	Point of measure	Tol in	XS in	S in	M in	L in	XL in	2XL in
A	Back length from HPS	±0.39	26.77	27.17	27.56	27.95	28.35	28.94
B	1/2 chest	±0.59	23.62	24.41	25.20	25.98	26.77	27.95
C	1/2 hem sweep, pleats closed	±0.79	24.02	24.80	25.59	26.38	27.17	28.35
CC	Hem circumference, pleats closed	±1.18	48.03	49.61	51.18	52.76	54.33	56.69
SH	Shoulder width (dropped)	±0.39	21.65	22.24	22.83	23.43	24.02	24.80
F	Armhole depth from HPS	±0.39	10.24	10.63	11.02	11.42	11.81	12.28
SL	Sleeve length from shoulder point	±0.39	21.65	22.05	22.44	22.83	23.23	23.62
I	1/2 bicep	±0.39	8.43	8.74	9.06	9.37	9.69	10.08
CW	Cuff, closed, laid flat	±0.20	4.61	4.76	4.92	5.08	5.24	5.43
CH	Cuff height	±0.12	2.36	2.36	2.36	2.36	2.36	2.36
N2	Collar stand length at neck seam	±0.39	17.17	17.64	18.11	18.58	19.06	19.65
N	Collar stand height at CB	±0.08	1.38	1.38	1.38	1.38	1.38	1.38
NF	Collar height at CB	±0.12	1.97	1.97	1.97	1.97	1.97	1.97
NP	Collar point length	±0.20	2.95	2.95	2.95	2.95	2.95	3.07
PW	Front placket width	±0.08	1.38	1.38	1.38	1.38	1.38	1.38
S1	First placket snap below HPS	±0.20	5.28	5.39	5.51	5.63	5.75	5.87
SN	Placket snap spacing	±0.20	4.72	4.82	4.92	5.02	5.12	5.24
YK	Back yoke depth at CB	±0.20	4.80	4.96	5.12	5.28	5.43	5.63
PS	Pleat seam from HPS	±0.39	12.36	12.68	12.99	13.31	13.62	14.02
PL	Pleated panel length	±0.39	14.41	14.49	14.57	14.65	14.72	14.92
P0	First pleat fold from CF / CB	±0.12	1.38	1.38	1.38	1.38	1.38	1.38
PF	Pleat face	±0.08	1.09	1.14	1.18	1.22	1.27	1.33
PD	Pleat depth	±0.12	2.36	2.36	2.36	2.36	2.36	2.36
PE	Pleat edge-stitch length	±0.08	0.79	0.79	0.79	0.79	0.79	0.79

INSIDE VIEW · overshirt (size M) · unlined, seams finished clean

back seen from the inside (as from the front) · the two fronts opened out at the sides · wearer's LEFT = image right

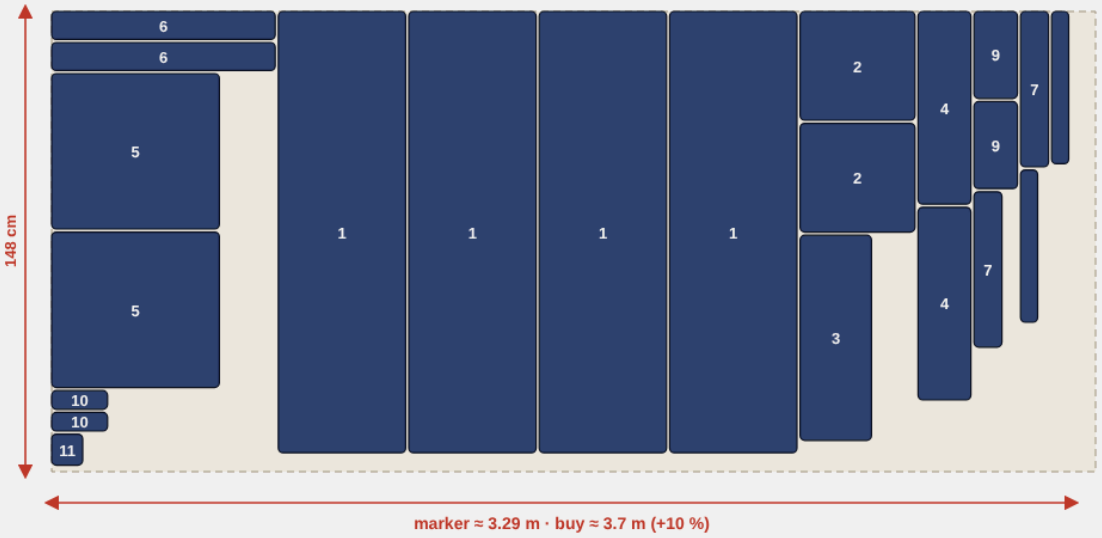


Item	Description	Weight / size	Colour	Qty (M)
Main fabric	100 % WOOL gabardine / fine twill, worsted, 220–240 g/m², NO polyester or other blend. Piece-dyed deep INDIGO with DISCHARGEABLE acid dyes (commissioned from PFD / ecru cloth; random stock indigo will not discharge: mill to confirm). No fluorocarbon / water-repellent, no heavy silicone softener, no resin or anti-felt finish (all block the dye): finish declaration required. Decatised. Discharge strike-off on the bulk lot before cutting	220–240 g/m² · 150 cm	Deep indigo (see colour standards)	marker ≈ 3.29 m + 10 % = BUY ≈ 3.7 m (M)
Hand-paint dyes	DISCHARGE paint (line item): rust + caramel acid dye on deep indigo would barely show (acid dye is transparent), so FP01 is DISCHARGE-PAINTED: a discharge paste = non-alkaline discharge agent for wool (Decrolin-type) + ILLUMINATING (discharge-resistant) acid dyes in RUST and CARAMEL, thickened with sodium alginate (Kemtex-type system, technical sheet DP35 'Discharge printing of wool dyed with acid dyes'); the paste strips the indigo and puts the colour in its place. Recipes from the approved lab dips. NOT pigment: no binder, the hand stays soft	≈ 6–10 g dye powder + ≈ 250 ml paste per set (estimate)	Rust + caramel	per set, incl. the trousers
Interfacing	Woven fusible for wool (e.g. Vlieseline G 785 or equal, low-temperature): collar + stand, cuffs, plackets, sleeve plackets. Fuse AFTER painting + steaming; run a steam test on fused collar / cuff / placket samples (30–60 min saturated steam can delaminate fusibles; also the press + dry-clean cycle)		Navy	≈ 0.5 m
Pleat-setting agent	Chemical permanent set for the knife pleats (monoethanolamine-sulphite type, Si-Ro-Set principle), applied by the pleating house before painting and again at the final re-set; SO_2 release: ventilated finishing house only			per pleating house recipe
Binding	Cupro bias 25 mm for Hong Kong finish on the pleat seam allowance and the pleated panel joins	25 mm	Indigo	≈ 4 m
Thread	Core-spun polyester (Amann Saba C 80 / Coats Epic 120 or equal) construction; Tex 40 topstitch. Polyester does NOT take acid dye or discharge: stitching sewn before painting (pleat seam, pleat edge-stitch) keeps its indigo colour inside the paint zones (accepted look) unless silk / polyamide thread is used for those rows (open question). Basting: 100 % polyester, pulled out after steaming		Indigo-matched (+ white basting)	≈ 450 m
Snaps	Ring-spring snap sets, cap + socket + stud + post, brass, NICKEL-FREE; see hardware	15 mm cap × 7, 12.5 mm × 1	A silver-tone (confirm) / B gunmetal	8 sets + 2 spare
Main label	ANIEZE woven damask, indigo-black ground, stone thread, real logo vector + PARIS; inner yoke at the inside back neck, centred, top edge 2 cm below the collar seam	7.5 × 4.2 cm	Black / stone	1
Size label	Woven, stone ground, black letter, under the main label	2.5 × 2 cm	Stone / black	1
Care + composition label	Printed satin, folded loop, wearer's LEFT side seam 10 cm above the pleat seam. Wording FR/EN (no symbols): DRY CLEAN ONLY, hand-painted, each piece unique	3 × 7 cm	Stone / black	1
Hanger loop	Self wool tube, 7 cm, under the collar at CB	7 cm	Indigo	1
Painting template	1:1 template per size (Mylar / heavy kraft): shape outlines with the ±2 cm envelope, CF / CB, pleat seam, shirt-hem and knee lines, pin points for the notches	per size		1 set per size (tooling)

Edge / seam	Material	Finish
Front plackets	wool, 2 plies + woven fusible	separate full-length strips, folded, edge-stitched 1.5 mm and topstitched 6 mm; run across the pleat seam
Collar + stand	wool + woven fusible	seamed, turned, edge-stitched 1.5 mm; stand topstitched 1.5 mm
Pleat seam	upper body over the pleated panel	allowances graded, Hong Kong bound with indigo cupro bias, pressed UP, topstitched 6 mm on the upper body
Pleated panel hem	wool	double-turned 1.5 cm and stitched BEFORE pleating; pleats pressed after painting
Side, armhole, sleeve seams	wool	flat-felled 6 mm, double topstitch (unlined, clean inside)
Yoke	wool, outer + inner	back and fronts sandwiched between the two yokes (burrito), topstitched 1.5 mm
Pleated panel joins (CB + side ends)	wool	plain 1 cm, Hong Kong bound, hidden in the inner fold of a pleat (CB join) / in the side seam
Cuffs	wool + woven fusible	folded cuff, edge-stitched 1.5 mm all round; 2 pleats in the sleeve into the cuff; tower placket 14 cm

FP01 OVERSHIRT · fabric marker estimate (indigo wool gabardine)

Shelf-packed bounding boxes (size M) on 148 cm usable width, grain along the length (↔ in this drawing). Real nested CAD markers are usually 10–20 % shorter.



Pieces (size M)

1. 4 × Pleated panel quarter (9 pleats, CF / CB end + side end, hem d… · 142 × 41
2. 2 × Upper front (HPS → pleat seam, dropped shoulder) · 35 × 37
3. 1 × Back body (yoke → pleat seam), on the fold · 66 × 23
4. 2 × Back yoke, outer + inner · 62 × 17
5. 2 × Sleeve · 50 × 54
6. 2 × Front placket strip (folded double) · 9 × 72
7. 2 × Collar, upper + under · 50 × 9
8. 2 × Collar stand, outer + inner · 49 × 5.5
9. 2 × Cuff (folded: outer + inner in one) · 28 × 14
10. 2 × Sleeve placket · 6 × 18
11. 1 × Hanger loop + small parts · 10 × 10

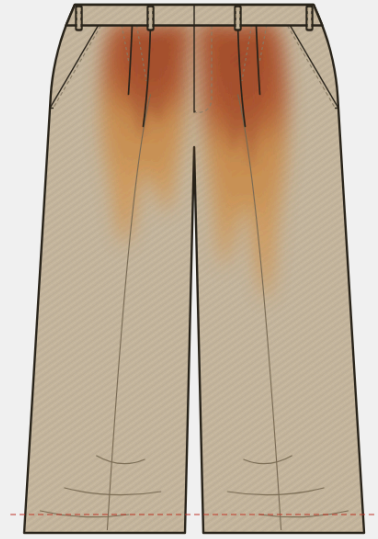
Net bounding-box area 3.87 m² · 230 g/m² cloth
Garment weight ≈ 0.79 kg at M (estimate)
Check: shrinkage after the full paint route (steam + rinse) on a 1 m test length BEFORE the marker is made.

Bounding-box estimate 3.29 m on 148 cm usable width (+10 % = 3.7 m). The 36 deep pleats need ≈ 560 cm of panel round: the pleated half is ≈ 5× its finished width. Sourcing estimate: 2.2–2.8 m; confirm with the factory marker.

1. Cloth: 100 % wool gabardine / twill 220–240 g/m², piece-dyed deep indigo, decatised. Cut with the grain vertical; pleated panel quarters cut across the full width (pleats run with the warp).
2. Silhouette: boxy unisex overshirt, hip length (A 70 cm at M), dropped shoulders (SH 58), relaxed chest (ease ≈ 20 cm), point collar on a stand, long sleeves with snap cuffs, back yoke. No pockets (open question).
3. Knife pleats all round the lower half: 9 per front half and 9 per back half = 36 pleats. Face PF ≈ 3 cm, depth PD 6 cm (intake 12 cm per pleat, 432 cm in all at M, so the panel is cut ≈ 560 cm round before pleating). Every fold turns TOWARD THE NEAREST SIDE SEAM: on the wearer's LEFT of CF the folds face the wearer's left side seam, on the wearer's RIGHT of CF they face the right side seam; the same on the back, so CB reads as a flat 7 cm panel between two pleats turning away from it (an inverted box look). First fold P0 = 3.5 cm from CF / CB, last fold ≈ 1.5 cm before the side seam. Each pleat is edge-stitched shut (1.5 mm from the fold) for the first PE = 2 cm below the pleat seam, then hangs free. Because PD > PF the underlays overlap: ≈ 5.0 layers sit under one point of the pleat seam at M, so the seam allowance is graded and Hong Kong bound.
4. Pleat seam: one level horizontal seam all round at PS = 33 cm below HPS at M (upper abdomen, ≈ 10 cm above the natural waist, as in the design image). Pleats are formed, basted and pressed first, the panel is joined to the upper body, allowances graded, Hong Kong bound, pressed UP and topstitched 6 mm on the upper body side. The front plackets are separate strips laid over it, so the seam stops at the placket.
5. THE PAINT is DISCHARGE painting (indigo removed and replaced by rust / caramel) and goes on BEFORE the rest of the garment is made (see the painting process, to be validated on proto panels): paint units = front sub-assembly (upper front + pleated front panel joined at PS; the plackets go on after painting and stay plain indigo) and back sub-assembly (back body + pleated back panel joined at PS, yoke NOT on yet). Pleats stay basted shut while painting, so only the pleat faces take colour and the underlays stay indigo: when the pleats open, indigo flashes between the painted faces (the 'fold' in fold-paint). Sleeves, yoke, collar, stand, cuffs stay plain indigo.
6. Sequence (to be validated on proto panels): cut → pleat the panels → chemical permanent set (MEA-sulphite type, Si-Ro-Set principle) → baste the pleats shut → join the paint units → discharge-paint on the registration template with the trouser panels → steam-fix → rinse / neutralise → release the basting → re-press / re-set the pleats → fuse the interfacings and finish the garment. Fallback B (paint flat, then pleat) breaks the composition differently (shapes split across faces and underlays) and is a fallback only.
7. Front: separate full-length plackets PW 3.5 cm, wearer's LEFT over right, from the neck to the hem across the pleat seam; 6 snaps on the front: 1 on the collar stand (12.5 mm) + 5 on the placket (15 mm) at S1 + k·SN. The pleats start P0 = 3.5 cm from CF.
8. Back: yoke (outer + inner) to YK at CB, back body sandwiched between the yokes, topstitched 1.5 mm; hanger loop under the collar. The back body below the yoke carries the paint, the yoke does not.
9. Collar: point collar on a stand, both interfaced with woven fusible, edge-stitched 1.5 mm. Stand snap 12.5 mm.
10. Sleeves: one-piece, set flat into the dropped armhole, flat-felled; 2 pleats into a 6 cm folded cuff with a 14 cm tower placket; 1 snap per cuff (15 mm).
11. Seams: flat-felled 6 mm on the armholes, side and sleeve seams (unlined garment, clean inside). Pleat seam and pleated panel joins: Hong Kong bound with indigo cupro bias.
12. Hem: the pleated panel hem is double-turned 1.5 cm and stitched before pleating (flat), so the hem folds crisply into each pleat.
13. Labels: main woven label (real ANIEZE logo vector, 7.5 × 4.2 cm) on the inner yoke at CB, top edge 2 cm below the collar seam; size label under it; care + composition in the wearer's LEFT side seam, 10 cm above the pleat seam. Care in words FR/EN: DRY CLEAN ONLY · hand-painted, each piece unique.
14. Machinery: single-needle lockstitch 3 mm stitch (topstitch 3.5 mm), felling machine or folder, steam press with pleat boards, snap press with the maker's dies. Test every seam, the pleat press and the snaps on a painted + steamed swatch first.
15. Care wording (FR/EN): 100 % LAINE / WOOL · NETTOYAGE À SEC UNIQUEMENT / DRY CLEAN ONLY · Ne pas laver, ne pas blanchir, ne pas sécher en tambour / Do not wash, bleach or tumble dry · Repasser à la vapeur sur l'envers avec une pattemouille, sans écraser les plis / Steam iron on the reverse with a press cloth, do not flatten the pleats · Peint à la main : chaque pièce est unique, les variations de couleur et de forme font partie du design / Hand-painted: each piece is unique; variations in colour and shape are part of the design.

FRONT

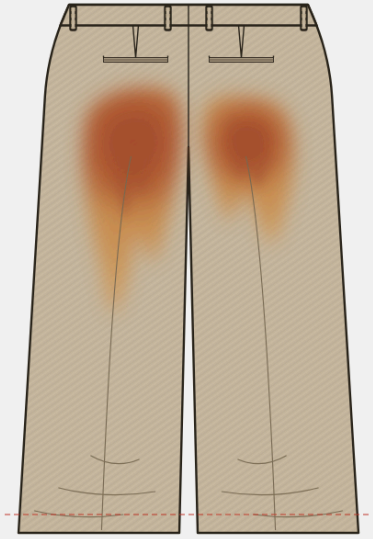
stone wool · 2 deep knife pleats per leg toward the side seam · permanent crease · slant pockets · paint over the pleats



laid flat · dashed line = the floor when worn: the last ≈4 cm pools in folds

BACK

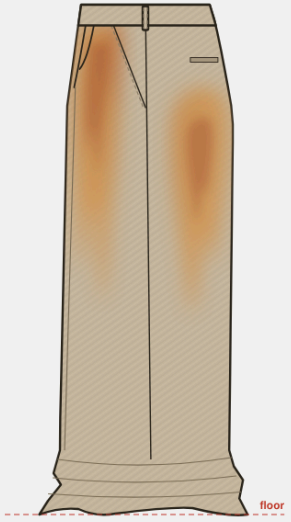
2 jetted pockets · darts · paint on the back thighs, ≥ 5 cm below the jets



laid flat · dashed line = the floor when worn: the last ≈4 cm pools in folds

LEFT SIDE (as worn)

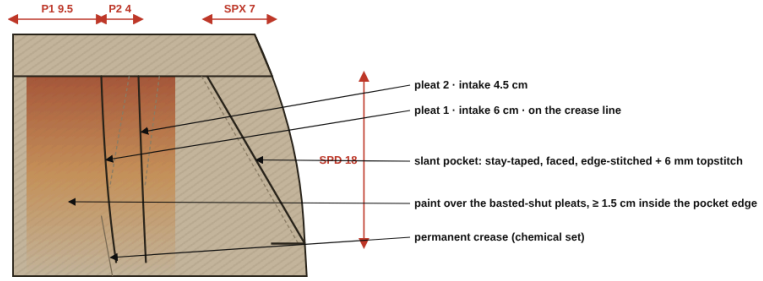
high rise · wide straight leg · hem pooling on the floor, nothing tied



Stone = wool gabardine · rust / caramel = hand-painted acid-dye composition continuing from the overshirt (see the paint map) · thin line = permanent crease · dashed = pleat underlay / topstitch

A · FRONT PLEATS + SLANT POCKET

Wearer's left front at M · 2 deep knife pleats toward the side seam · paint over the pleats



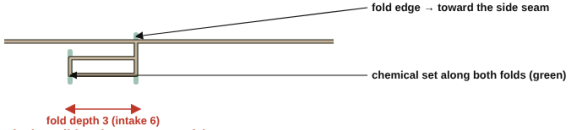
Folds turned toward the side seam, open below the waistband (permanent set). Waistband, loops and pocket parts stay plain stone.

B · PERMANENT PLEAT + CREASE (Si-Ro-Set principle)

Chemical set before painting, re-set after · to be validated on proto panels

stone wool gabardine 300 g/m² · woven fusible · pocketing · cupro lining

① knife pleat section at the waistband seam (schematic)



② sequence (to be validated on proto panels)

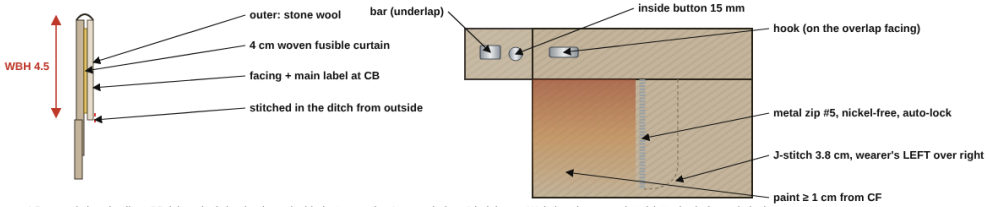
1. Cut, mark the pleat and crease lines (thread-trace).
2. Pleating house: fold the 2 pleats + front and back creases, apply the MEA-sulphite set agent, press / steam.
3. Baste the pleats shut (polyester thread, 1 cm from the folds) for painting.
4. Paint, steam-fix 102 °C 30–45 min, rinse ≤ 40 °C without agitation.
5. Release the basting, re-press and RE-SET the pleats + creases (second set pass).

Steam alone gives a cohesive set that steam and water release: the chemical set makes it permanent.

C · WAISTBAND + FLY

4.5 cm finished · hook + bar + inside button · metal zip #5

stone wool gabardine 300 g/m² · woven fusible · pocketing · cupro lining

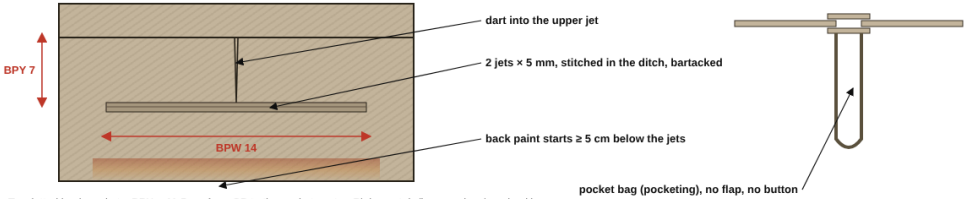


4.5 cm waistband split at CB (alteration), hook + bar + inside button on the 4 cm underlap; 6 belt loops. Waistband never painted (attached after painting).

D · BACK JETTED POCKET

2 jets × 5 mm · dart · bag in pocketing

stone wool gabardine 300 g/m² · woven fusible · pocketing · cupro lining

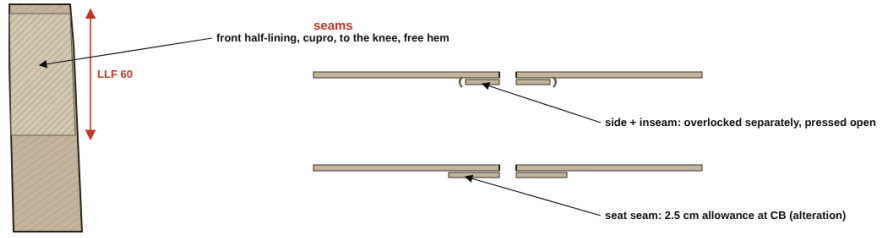


Two jetted back pockets, BPX = 11.5 cm from CB to the pocket centre. Plain, no tab (house rule: clean back).

E · FRONT HALF-LINING + SEAMS

Cupro, front only, to the knee · overlocked seams

stone wool gabardine 300 g/m² · woven fusible · pocketing · cupro lining

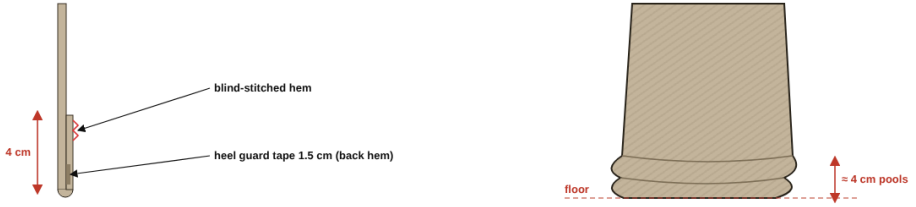


Back unlined. The lining is joined at the waistband seam, round the fly and into the side seams to the pocket bag; never painted, attached after painting.

F · HEM + POOLING

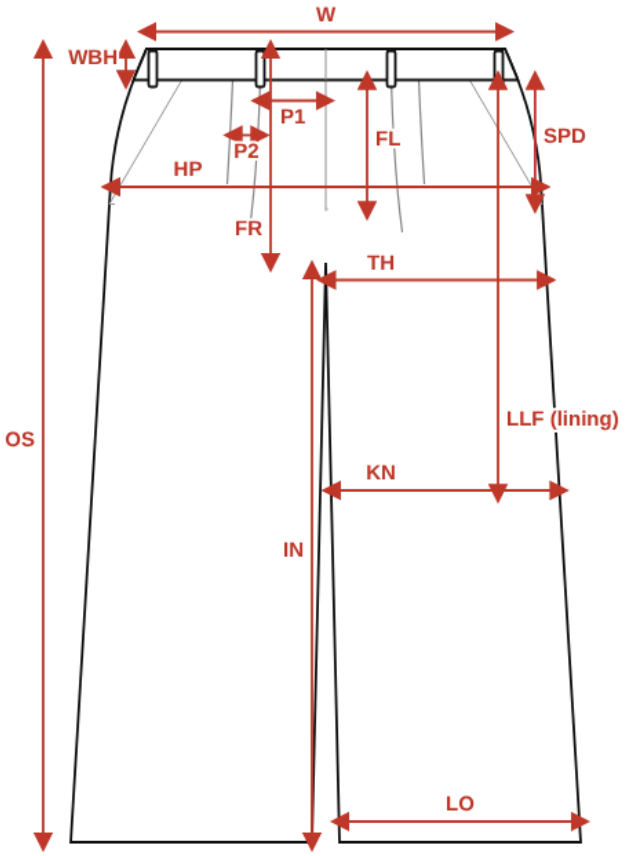
Turned 4 cm, blind-stitched · heel tape · ≈ 4 cm pooling

stone wool gabardine 300 g/m² · woven fusible · pocketing · cupro lining

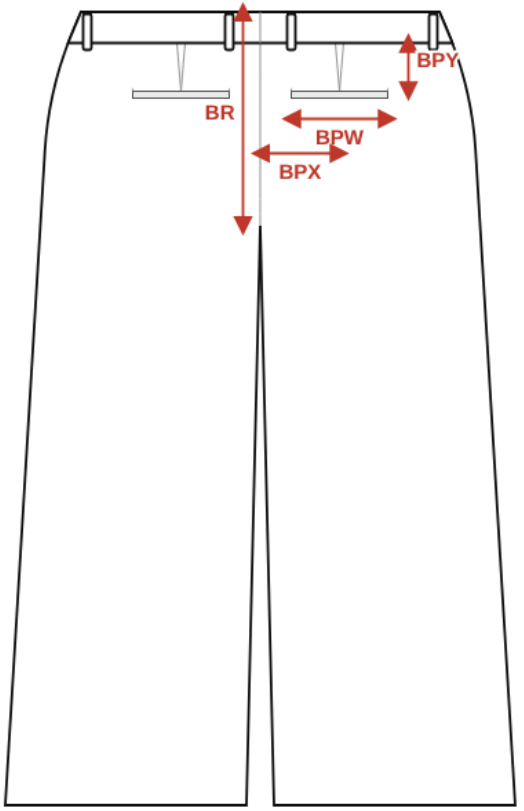


Pooling is in the length (inseam 84 at M). No turn-up, nothing tied at the leg. The crease (chemical set) runs into the pooled folds.

FRONT · points of measure (M)



BACK



Inseam 84 cm at M (174 cm, worn to the floor): body crotch height ≈ 80 cm; a wide high-rise trouser hangs its crotch ≈ 2 cm below the body crotch, leaving ≈ 78 cm to the floor barefoot; + ≈ 2 cm shoe sole = 80 cm in shoes; + ≈ 4 cm of pooled excess folding on the floor = 84 cm. Graded +1 cm per size with the crotch height, so every size pools the same ≈ 4 cm (house rule: wide, high-rise, pooling, nothing tied at the leg).

Code	Point of measure	How to measure	Tol	Grade	XL→2XL	XS	S	M	L	XL	2XL
W	1/2 waist, relaxed	Top edge of the waistband, closed (hook + bar and inside button done up), laid flat, edge to edge	±1	2	3	48	50	52	54	56	59
WBH	Waistband height, finished	Top edge to the waistband seam, at CF and CB	±0.2	0	0	4.5	4.5	4.5	4.5	4.5	4.5
HP	1/2 hip	Laid flat, pleats closed, side to side 20 cm below the top of the waistband (+0.5 cm per size)	±1.5	2	2.5	58.5	60.5	62.5	64.5	66.5	69
TH	1/2 thigh	One leg laid flat, 2.5 cm below the crotch point, perpendicular to the crease, fold to fold	±1	1.2	1.5	37.6	38.8	40	41.2	42.4	43.9
KN	1/2 knee	One leg laid flat, 33 cm below the crotch point measured along the inseam, fold to fold	±1	0.8	1	35.4	36.2	37	37.8	38.6	39.6
LO	1/2 leg opening	One leg laid flat, straight across the hem edge	±1	0.6	0.8	33.8	34.4	35	35.6	36.2	37
LOC	Hem circumference (leg opening, full)	One leg, all the way round the hem edge (= 2 × LO)	±1.5	1.2	1.6	67.6	68.8	70	71.2	72.4	74
FR	Front rise incl. waistband	Top of the waistband at CF, along the crotch seam curve to the crotch point (zip closed)	±0.7	0.7	1	32.6	33.3	34	34.7	35.4	36.4
BR	Back rise incl. waistband	Top of the waistband at CB, along the seat seam curve to the crotch point	±0.7	1	1.3	42	43	44	45	46	47.3
IN	Inseam	Crotch point straight down the inside leg seam to the hem edge	±1	1	1	82	83	84	85	86	87
OS	Outseam incl. waistband	Top of the waistband at the side seam, along the side seam to the hem edge	±1.5	1.6	2	112.8	114.4	116	117.6	119.2	121.2
P1	Pleat 1 position from CF	Along the waistband seam, CF to the fold of pleat 1 (on the crease line)	±0.3	0.4	0.5	8.7	9.1	9.5	9.9	10.3	10.8
P2	Pleat spacing	Along the waistband seam, fold of pleat 1 to fold of pleat 2	±0.3	0.1	0.2	3.8	3.9	4	4.1	4.2	4.4
PD1	Pleat 1 intake	Total cloth taken in by pleat 1 at the waistband seam (fold depth x 2)	±0.3	0	0	6	6	6	6	6	6
PD2	Pleat 2 intake	Total cloth taken in by pleat 2 at the waistband seam (fold depth x 2)	±0.3	0	0	4.5	4.5	4.5	4.5	4.5	4.5
SPX	Slant pocket, top from side seam	Along the waistband seam, side seam to the top corner of the pocket edge	±0.3	0.2	0.3	6.6	6.8	7	7.2	7.4	7.7
SPD	Slant pocket depth at side seam	Waistband seam straight down the side seam to the lower bartack of the pocket opening	±0.5	0.3	0.5	17.4	17.7	18	18.3	18.6	19.1
BPY	Back pocket below the waistband seam	Waistband seam straight down to the top edge of the upper jet	±0.3	0.2	0.3	6.6	6.8	7	7.2	7.4	7.7
BPW	Back pocket opening width	Jet end to jet end (bartack to bartack)	±0.3	0.3	0.5	13.4	13.7	14	14.3	14.6	15.1
BPX	Back pocket centre from CB seam	Along the waistband seam, CB seam to the centre of the pocket opening	±0.3	0.5	0.6	10.5	11	11.5	12	12.5	13.1
FL	Fly opening length	Waistband seam to the J-stitch bartack at CF	±0.5	0.5	0.5	18	18.5	19	19.5	20	20.5
LLF	Front half-lining length	Waistband seam straight down the front, parallel to the crease, to the lining hem edge (ends at the knee)	±1	1	1	58	59	60	61	62	63
BL	Belt loop length, finished	Waistband seam to the top fold of the loop	±0.2	0	0	5.5	5.5	5.5	5.5	5.5	5.5

Fit check (cm)	Target	XS	S	M	L	XL	2XL
Waist ease (2W − body waist)	1.5–3.5	2.2	2.2	2.2	2.2	2.2	2.2
Hip ease (2HP − body hip)	16–20	18.1	18.1	18.1	18.1	18.1	18.1
Pooling (IN − body crotch height)	2–6	4.0	4.0	4.0	4.0	4.0	4.0
Outseam − inseam	29–35	30.8	31.4	32.0	32.6	33.2	34.2
Back rise − front rise	8–12	9.4	9.7	10.0	10.3	10.6	10.9

Code	Point of measure	Tol in	XS in	S in	M in	L in	XL in	2XL in
W	1/2 waist, relaxed	±0.39	18.90	19.69	20.47	21.26	22.05	23.23
WBH	Waistband height, finished	±0.08	1.77	1.77	1.77	1.77	1.77	1.77
HP	1/2 hip	±0.59	23.03	23.82	24.61	25.39	26.18	27.17
TH	1/2 thigh	±0.39	14.80	15.28	15.75	16.22	16.69	17.28
KN	1/2 knee	±0.39	13.94	14.25	14.57	14.88	15.20	15.59
LO	1/2 leg opening	±0.39	13.31	13.54	13.78	14.02	14.25	14.57
LOC	Hem circumference (leg opening, full)	±0.59	26.61	27.09	27.56	28.03	28.50	29.13
FR	Front rise incl. waistband	±0.28	12.83	13.11	13.39	13.66	13.94	14.33
BR	Back rise incl. waistband	±0.28	16.54	16.93	17.32	17.72	18.11	18.62
IN	Inseam	±0.39	32.28	32.68	33.07	33.46	33.86	34.25
OS	Outseam incl. waistband	±0.59	44.41	45.04	45.67	46.30	46.93	47.72
P1	Pleat 1 position from CF	±0.12	3.43	3.58	3.74	3.90	4.06	4.25
P2	Pleat spacing	±0.12	1.50	1.54	1.57	1.61	1.65	1.73
PD1	Pleat 1 intake	±0.12	2.36	2.36	2.36	2.36	2.36	2.36
PD2	Pleat 2 intake	±0.12	1.77	1.77	1.77	1.77	1.77	1.77
SPX	Slant pocket, top from side seam	±0.12	2.60	2.68	2.76	2.83	2.91	3.03
SPD	Slant pocket depth at side seam	±0.20	6.85	6.97	7.09	7.20	7.32	7.52
BPY	Back pocket below the waistband seam	±0.12	2.60	2.68	2.76	2.83	2.91	3.03
BPW	Back pocket opening width	±0.12	5.28	5.39	5.51	5.63	5.75	5.94
BPX	Back pocket centre from CB seam	±0.12	4.13	4.33	4.53	4.72	4.92	5.16
FL	Fly opening length	±0.20	7.09	7.28	7.48	7.68	7.87	8.07
LLF	Front half-lining length	±0.39	22.83	23.23	23.62	24.02	24.41	24.80
BL	Belt loop length, finished	±0.08	2.17	2.17	2.17	2.17	2.17	2.17

INSIDE VIEW · trouser (size M) · front half and back half seen from the inside

cut open at the side seams, seen as from the front: the wearer's LEFT is on the image right · lining and labels go on after painting + steaming

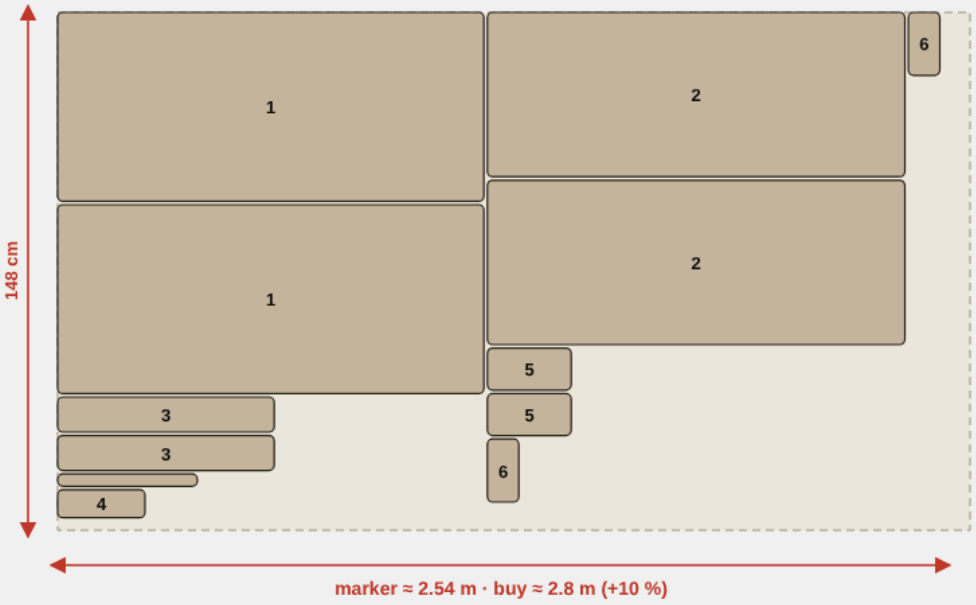


Item	Description	Weight / size	Colour	Qty (M)
Main fabric	100 % WOOL gabardine, worsted, 280–320 g/m², NO polyester; warm STONE, piece-dyed with acid dyes (or PFD dyed to shade); no fluorocarbon / water-repellent, no heavy silicone, no resin / anti-felt finish (they block the paint): finish declaration required; decatised	280–320 g/m² · 150 cm	Warm stone (see colour standards)	marker ≈ 2.54 m + 10 % = BUY ≈ 2.8 m (M)
Hand-paint dyes	Hand-paint dyes (line item): DIRECT acid-dye painting on the light stone base: milling or 1:2 metal-complex acid dyes (Lanaset / Dharmaset type, better wet, dry-clean and light fastness than levelling acid dyes) in RUST and CARAMEL, thickened with sodium alginate for brushing, diluted for spraying; citric acid / ammonium sulphate to pH 4–5. Shade-matched to the overshirt's illuminating discharge colours in ONE lab-dip round. NOT pigment: no binder	≈ 4–6 g dye powder + ≈ 200 ml paste per pair (estimate)	Rust + caramel	per pair
Pleat-setting agent	Chemical permanent set (MEA-sulphite type, Si-Ro-Set principle) for the 2 front pleats per leg + front and back creases; at the pleating house (SO_2: ventilation + PPE)			per pleating house recipe
Lining (front half)	Bemberg cupro twill, front only, waistband seam to the knee (LLF), free hem; stone-matched	≈ 75–85 g/m², 140 cm	Stone	≈ 0.8 m (M)
Pocketing	Cotton-cupro pocketing twill: slant pocket bags, back pocket bags, waistband facing backing	≈ 110 g/m², 145 cm	Stone	≈ 0.5 m (M)
Interfacing	Woven fusible (low temperature) on the waistband (4 cm curtain), fly facing + shield, pocket edges, jet boxes; fuse AFTER painting + steaming and steam-test fused samples		Stone	≈ 0.4 m
Stay tape + heel tape	10 mm cotton stay tape on the slant pocket edges; 1.5 cm heel guard tape inside the back hem	10 / 15 mm	Stone	≈ 0.6 m + 0.8 m
Thread	Core-spun polyester construction + topstitch (does not take acid dye: stitching sewn before painting stays stone inside the paint zones, accepted); polyester basting thread for the paint basting		Stone-matched	≈ 300 m
Zip, fly	Metal zip #5, closed end, auto-lock slider, nickel-free brass teeth; tape stone; plain puller (no logo)	18 cm (XS–S 17, M–L 18, XL–2XL 19)	A silver-tone / B gunmetal, matching the snaps	1
Hook and bar	Flat trouser hook + bar, brass, nickel-free, 4 prongs each	≈ 25 mm hook	A / B	1 set
Inside button	Flat 4-hole button, brass, nickel-free, on the underlap extension	15 mm	A / B	1 + 1 spare
Main label	The ANIEZE woven label (real logo vector + PARIS), scaled to fit the 4.5 cm waistband: on the waistband facing at CB, sewn 4 sides	6 × 3.4 cm	Black / stone	1
Size label	Woven, stone ground, black letter, in the wearer's LEFT side seam above the care label	2.5 × 2 cm	Stone / black	1
Care + composition label	Printed satin, folded loop, wearer's LEFT side seam, 12 cm below the waistband. Wording FR/EN (no symbols): DRY CLEAN ONLY, hand-painted, each piece unique, permanent pleats	3 × 7 cm	Stone / black	1

Edge / seam	Material	Finish
Side seams + inseam	wool	plain 1 cm, overlocked separately in stone, pressed open
Seat seam	wool	2.5 cm allowance at CB (alteration), double-stitched, overlocked
Slant pocket edges	wool + stay tape	stay-taped, faced, understitched, edge-stitched 1.5 mm + topstitched 6 mm
Back pocket jets	wool	2 jets × 5 mm, stitched in the ditch, bartacked
Waistband	wool + woven curtain	outer + facing on a 4 cm woven fusible curtain, top edge seamed + understitched; split at CB
Hem	wool	turned 4 cm, blind-stitched; 1.5 cm heel guard tape inside the back hem (the hem pools)
Front half-lining	cupro	joined at the waistband seam, fly and side seams; free 1 cm double-turned hem at the knee
Belt loops	wool, folded 3-ply	edge-stitched 1.5 mm both sides; bartacked under the waistband top edge

FP02 TROUSER · fabric marker estimate (stone wool gabardine)

Shelf-packed bounding boxes (size M) on 148 cm usable width, grain along the length (↔ in this drawing). Real nested CAD markers are usually 10–20 % shorter.



Pieces (size M)

1. 2 × Back leg (incl. seat, CB allowance 2.5 cm, hem 4 cm) · 54 × 122
2. 2 × Front leg (incl. 2 pleats, grown-on fly facing on the wearer's… · 47 × 119.5
3. 2 × Waistband outer + facing (split at CB) · 10 × 62
4. 1 × Fly shield (right) · 8 × 25
5. 2 × Slant pocket facing · 12 × 24
6. 2 × Back pocket jets + facings · 18 × 9
7. 1 × Belt loop strip · 3.5 × 40

Net bounding-box area 2.69 m² · 300 g/m² cloth
Garment weight ≈ 0.83 kg at M (estimate)
Check: shrinkage after the full paint route (steam + rinse) on a 1 m test length BEFORE the marker is made.

Bounding-box estimate 2.54 m (+10 % = 2.8 m). The very wide legs (½ opening 35 cm) do not fit four across 150 cm. Sourcing estimate: 1.6–2.0 m; confirm with a nested CAD marker.

1. Cloth: 100 % wool gabardine 280–320 g/m², warm stone, no finish that blocks dye. Cut with the crease on the straight grain; all pieces of one pair from one lot.
2. Fit: very wide, high rise, unisex (house rule: wide, high-rise, pooling on the floor, nothing tied at the leg). Inseam 84 cm at M (174 cm, worn to the floor): body crotch height ≈ 80 cm; a wide high-rise trouser hangs its crotch ≈ 2 cm below the body crotch, leaving ≈ 78 cm to the floor barefoot; + ≈ 2 cm shoe sole = 80 cm in shoes; + ≈ 4 cm of pooled excess folding on the floor = 84 cm. Graded +1 cm per size with the crotch height, so every size pools the same ≈ 4 cm (house rule: wide, high-rise, pooling, nothing tied at the leg).
3. Front: two deep knife pleats per leg at the waistband seam, folds turned toward the SIDE seam (P1 on the crease line, intake 6 cm; P2 4 cm further out, intake 4.5 cm), open below the waistband; pleat 1 runs into the front crease to the hem. Permanent pleats + crease: the two front knife pleats and the front and back creases get a CHEMICAL permanent set (monoethanolamine-sulphite type, Si-Ro-Set principle) at the pleating / finishing house, before painting; after painting, steaming and rinsing they are re-pressed and re-set. Steam alone gives a 'cohesive' set that water and steam release. To be validated on proto panels.
4. THE PAINT: direct acid-dye painting (rust + caramel on the light stone base) on the front legs (over the basted-shut pleats, from the waistband seam down, fading out between mid-thigh and knee) and on the back thighs (from below the back pockets, fading out at mid-thigh to knee). The legs are painted on the registration template together with the overshirt's paint units so the composition continues from the shirt's pleats onto the trouser. Waistband, loops, fly shield, pocket parts stay plain stone. See the paint map + painting process.
5. Slant front pockets: opening from SPX at the waistband seam to SPD down the side seam; stay-taped, faced, understitched, edge-stitched + topstitched 6 mm; cotton-cupro bags caught in the fly (extended stay). Paint stays ≥ 1.5 cm inside the pocket edge.
6. Fly: metal #5 zip, wearer's LEFT over right, grown-on fly facing on the left front, separate right fly shield, J-stitch 3.8 cm. Waistband: 4.5 cm, outer + facing on a woven curtain, split at CB for alteration; hook + bar + inside button on the 4 cm underlap.
7. Belt loops: 6 (front on the pleat-1 line, at the side seams, back 4.5 cm from CB), folded 3-ply, edge-stitched, caught in the waistband seam and bartacked under the top edge.
8. Back: one dart per side into the jet; two jetted pockets (2 × 5 mm jets), no flap or button. The back paint starts ≥ 5 cm below the jets.
9. Front half-lining: Bemberg cupro, front only, from the waistband seam to the knee (LLF), joined at the waistband seam, round the fly and into the side seams down to the pocket bag; free 1 cm double-turned hem. Back unlined, seams overlocked.
10. Hem: turned 4 cm, blind-stitched, heel guard tape inside the back; the hem pools ≈ 4 cm on the floor in soft folds, no turn-up.
11. Labels: main woven label (real ANIEZE logo vector, 6 × 3.4 cm) on the waistband facing at CB; size + care in the wearer's LEFT side seam 12 cm below the waistband. Care in words FR/EN: DRY CLEAN ONLY · hand-painted, each piece unique · permanent pleats.
12. Machinery: single-needle lockstitch 3 mm, overlocker, blind-hem machine, bartack, pleating / crease press with the set agent at the finishing house, steam press with a pressing cloth.
13. Care wording (FR/EN): Corps / Shell : 100 % laine / wool · Doublure / Lining (devant / front) : 100 % cupro · NETTOYAGE À SEC UNIQUEMENT / DRY CLEAN ONLY · Ne pas laver, ne pas blanchir, ne pas sécher en tambour / Do not wash, bleach or tumble dry · Plis permanents : repasser à la vapeur avec une pattemouille, dans les plis / Permanent pleats: steam with a press cloth along the pleats · Peint à la main : chaque pièce est unique / Hand-painted: each piece is unique.

A · MAIN LABEL (overshirt inside back neck · trouser waistband)





trouser: 6 × 3.4 cm (fits the 4.5 cm band)

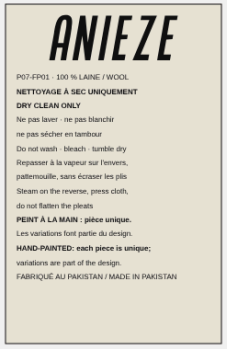
Woven damask, black ground, stone thread · ANIEZE wordmark from the master vector · PARIS at 22 % of the logo height · margin 0.45 cm
Overshirt: inner yoke, centred at CB, top edge 2 cm below the collar seam · Trouser: waistband facing at CB, sewn 4 sides · added AFTER painting + steaming

B · SIZE LABEL

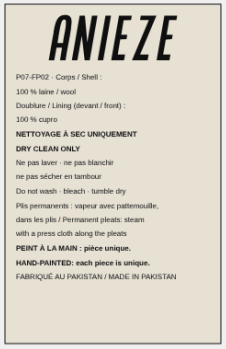


Woven, stone ground, black letter, 2.5 × 2 cm · overshirt: under the main label · trouser: left side seam above the care label

C · CARE + COMPOSITION (wording FR / EN, no symbols)



overshirt: LEFT side seam,
10 cm above the pleat seam



trouser: LEFT side seam,
12 cm below the waistband

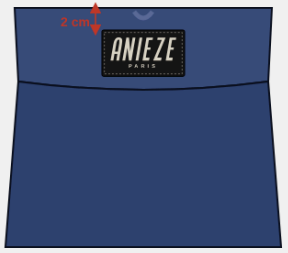
Printed satin 3 × 7 cm (drawn 1.6×), folded loop.
DRY CLEAN ONLY + hand-painted / each piece unique, in FR and EN.
No ISO 3758 symbols needed (founder); wording to be checked by GINETEX France or a lawyer before bulk.
Labels are polyester: sewn on after painting (they do not take dye).

D · LOGO MASTER



Use only this vector
(Drive: Logo black.svg)
No outlines, no effects,
no redrawn letters

E · WHERE THE BRANDING SITS



Overshirt: inner yoke, main label centred at CB,
top edge 2 cm below the collar seam; size label under it.

Hanger loop: self wool, 7 cm, under the collar.

Care + composition: LEFT side seams (both pieces).



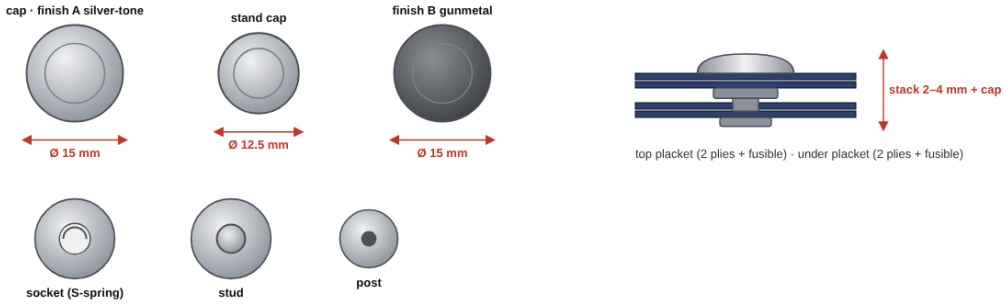
trouser waistband facing, CB (split seam)

Trouser: main label 6 × 3.4 cm on the
waistband facing at CB, beside the split seam.

Logo from the ANIEZE master vector only (brand/anieze_logo_black.svg), never typed. Overshirt main label 7.5 × 4.2 cm on the inner yoke, 2 cm below the collar seam, centred at CB; trouser 6 × 3.4 cm on the waistband facing at CB. Care in words FR/EN: DRY CLEAN ONLY · hand-painted, each piece unique.

A · RING-SPRING SNAP (overshirt)

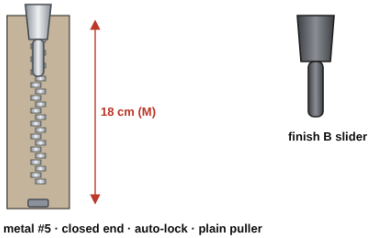
8 sets: stand 1 × 12.5 mm, placket 5 + cuffs 2 × 15 mm · brass, nickel-free · drawn at 7 × scale



- Cap + socket on the top (wearer's LEFT) placket, stud + post on the under placket; cuff: cap on the overlap. Fusible patch behind each.
- Set with the maker's die AFTER painting, steaming and pressing (no metal in the steamer). Open / close 15–30 N, pull-off ≥ 90 N on the real stack.
- Plain cap face (no logo deboss) unless the founder asks; a logo would use the real ANIEZE vector only.

B · FLY ZIP (trouser)

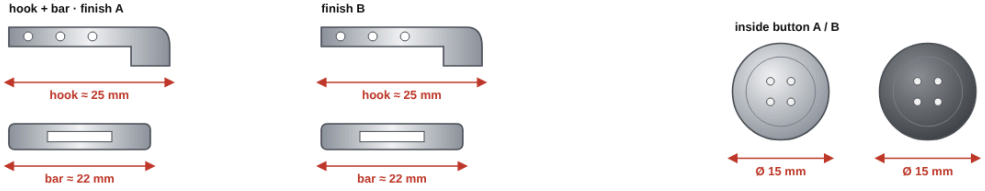
Metal #5, brass teeth, nickel-free, closed end, auto-lock slider, stone tape



- Length by size: XS–S 17 cm, M–L 18 cm, XL–2XL 19 cm (fly opening FL − 1).
- Wearer's LEFT over right, J-stitch 3.8 cm. EN 16732 zip strength.
- Tape colour stone; teeth + slider in the SAME finish as the snaps.
- Sewn in after painting (the fly facing is painted, the zip is not).

C · HOOK + BAR, INSIDE BUTTON (trouser)

Flat trouser hook + bar, brass, nickel-free · 15 mm flat 4-hole button, brass



- Hook on the overlap facing (wearer's left), bar on the underlap; 4 prongs each through the waistband + curtain, washer plates inside.
- Inside button on the 4 cm underlap extension, buttonhole in the overlap facing; sewn with a thread shank. Pull ≥ 90 N.
- Same finish lot as the overshirt snaps (one lot per set).

D · FINISHES + TESTS

All metal, brass, NICKEL-FREE · one finish lot per set (shirt + trouser)

A · nickel-free SILVER-TONE (render; confirm)
PVD, or tin-based plating (e.g. white bronze). NOT nickel plating.

B · nickel-free GUNMETAL (option)
PVD or tin-cobalt. Not black nickel.

- EN 1811 nickel release AFTER EN 12472 wear simulation, for EACH finish.
- REACH Annex XVII: Pb < 0.05 % (63) · Cd < 0.01 % (23) · Ni < 0.5 µg/cm²/week (27).
- ISO 9227 neutral salt spray 24 h, no red rust.
- One dry-clean cycle (perc / hydrocarbon) on a hardware card: no finish loss.
- Pull tests ≥ 90 N on the real wool + fusible stack, 5 of each at proto.
- Hardware never goes through the steamer or the paint (set after).
- Approve one hardware card per finish before bulk.

Item	Specification	Size	Finish	Qty	Where	Checks
Snap, front + cuffs	Ring-spring (S-spring) snap: cap + socket on the top layer, stud + post below, brass, set with the maker's die through 3–4 plies of wool + fusible	cap Ø 15 mm, post for 2–4 mm stack	A: nickel-free SILVER-TONE (PVD, or tin-based plating e.g. white bronze) · B: nickel-free gunmetal	7	5 placket (S1 + k·SN), 1 per cuff	Pull-off ≥ 90 N; open / close force 15–30 N; EN 1811 after EN 12472
Snap, collar stand	Same snap, smaller cap	cap Ø 12.5 mm	A / B as above	1	Collar stand, centred on the stand height	Pull-off ≥ 70 N; EN 1811
Fly zip	Metal #5 closed-end, auto-lock slider, nickel-free brass teeth, tape stone	17 / 18 / 19 cm	A silver-tone / B gunmetal	1	CF fly, wearer's LEFT over right, J-stitch 3.8 cm	EN 16732 zip strength; EN 1811 nickel release
Hook and bar	Flat trouser hook (on the overlap facing) + bar (on the underlap), brass, nickel-free	25 mm	A / B	1 set	Waistband at CF	Pull ≥ 90 N; EN 1811
Inside button	Flat 4-hole button, brass, nickel-free, sewn with a thread shank	15 mm	A / B	1	Underlap extension, buttoning into the overlap facing	Pull ≥ 90 N; EN 1811

All hardware METAL, brass, one finish lot per set (shirt + trouser): A = NICKEL-FREE silver-tone (the design image shows bright silver: PVD, or tin-based plating such as white bronze, NOT nickel plating; confirm) · B = nickel-free gunmetal (PVD or tin-cobalt). EN 1811 nickel release tested AFTER EN 12472 wear/corrosion simulation for EACH finish; REACH Annex XVII: Pb < 0.05 % (entry 63), Cd < 0.01 % (entry 23), Ni release < 0.5 µg/cm²/week (entry 27); 24 h neutral salt spray (ISO 9227). Hardware goes on AFTER painting, steaming and pressing (never steamed with acid dye). Pull tests on the real wool + fusible stack, 5 of each at proto. Dry-clean resistance: one cycle in perchloroethylene / hydrocarbon on a hardware card, no finish loss.

ANIEZE: sourcing research for the painted, pleated wool gabardine set

Garments: an INDIGO wool gabardine overshirt (220–240 g/m²) with a knife-pleated lower half, and STONE wool gabardine wide trousers (280–320 g/m²) with permanent knife pleats. Both are hand-painted with acid dyes (rust and caramel) and steam-fixed. **Production:** Sialkot, Pakistan. Runs of about 50–300 sets. **Research date:** 11 Nov 2026. This was desk research only. No supplier was contacted.

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0. How to read this document (verification is limited)

- **The research environment blocked direct page fetches.** Pages on nona-source.com, jacquardproducts.com and pnac.gov.pk could not be opened. Everything below comes from web-search result summaries and snippets.
- **"Verified (snippet)"** means a search result from the supplier's own site, or from an official registry such as PNAC, stated the fact. I did not open the page itself, so re-check prices, MOQs and accreditation scopes before relying on them.
- **"Secondary"** means a retailer, directory, news article or trade-fair listing stated it.
- **"Unverified"** means my own trade knowledge or an inference. It is a lead to check, not a fact.
- **No published MOQ was found** for VBC, Marzotto, Reda, Drago, Loro Piana, Yünsa, Altınyıldız, Holland & Sherry or Marling & Evans. These mills give terms to trade accounts on request.
- **No Pakistani worsted gabardine weaver was verified,** and neither was any Pakistani pleating house, wool garment-dye or steam house, or acid-dye hand-painting studio. Section 6 lists the gaps.

Rough fabric quantities (my estimate, not a supplier figure)

These assume 150 cm cloth.

Garment	Approx. use per piece	50 pieces	300 pieces
Overshirt (pleated lower half takes about 3× width in the pleated zone)	2.2–2.8 m	~130 m indigo	~800 m indigo
Wide trousers with pleats	1.6–2.0 m	~90 m stone	~550 m stone

Add 5–10% for painting trials, pleat trials and lab samples. A 50-piece run is below most mill minimums, so deadstock and stock platforms fit it best. A 300-piece run may reach mill or stock-service minimums.

Weight check

Ask every supplier whether its weight is **g/m² or g per linear metre**. At 150 cm width, 1 g/m² equals 1.5 g per linear metre.

Garment	Target g/m²	Same target in g per linear metre at 150 cm
Overshirt	220–240 g/m²	330–360 g/lm
Trousers	280–320 g/m²	420–480 g/lm

Two listings below are ambiguous on this:

- Zerow's "271 g/m" would be only about 181 g/m² if it is per linear metre.
- Joel & Son's "350 g" would be about 233 g/m² if it is per linear metre, which fits the overshirt.

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1. Process risks to settle before buying anything

1. **Rust and caramel acid dye painted on an indigo or navy base will hardly show.** Acid dye is transparent, so it reads as a muddy darkening on a dark ground. There are three ways around this:
   - **(a) Accept a tonal effect.** Only try this after a test.
   - **(b) Use discharge painting.** The base cloth must be dyed with *dischargeable* acid dyes. You then paint a discharge paste containing *illuminating* (discharge-resistant) acid dyes. Kemtex (UK) sells all three components: dischargeable acid dyes, illuminating acid dyes, and Decrolin discharge agent. Kemtex says Decrolin is preferred for wool because it is not alkaline, and has a technical sheet, DP35 "Discharge Printing of Wool Dyed with Acid Dyes". This means you must commission the base dyeing (from PFD/ecru cloth, or with a mill), because random stock indigo cloth will not be dischargeable.
   - **(c) Use a lighter "indigo" base shade.** The stone base is fine for rust and caramel.
2. **Order of operations.** Wool pleats set with steam alone are a "cohesive set": water or steam can release them. A permanent pleat in wool needs a chemical setting agent, which is the Si-Ro-Set idea. In practice:
   - Paint, steam-fix and wash off the pleat panels **flat** first.
   - Then pleat them with a chemical-assisted set.
   - If you steam-fix a garment that is already pleated, expect the pleats to move. Trial both routes.
3. **The base cloth must be dyeable.** Ask for:
   - no fluorocarbon or other water-repellent finish;
   - no heavy silicone softener;
   - no resin or anti-felt finish that blocks acid dye (or a declaration of what finish is applied). PFD (prepared for dyeing) or "ready-to-dye" cloth is ideal.
4. **Fibre blends.** Polyester in a blend does not take acid dye, so those areas stay unpainted and the colour looks heathered. Nylon/polyamide and silk do take acid dye. Elastane is fine in small amounts.
5. **Thread and trims.** Polyester core-spun thread (for example Coats Epic) will not take acid dye. Neither will polyester woven labels or polyester interlinings. If paint crosses topstitching, choose the thread colour on purpose, or use silk or polyamide thread.
6. **Fusing and steaming.** 30–60 minutes of saturated steam at about 100–104 °C can delaminate some fusibles. Paint and steam before fusing, or test fused panels through the full steam cycle.
7. **Wash-off decides the rubbing result (ISO 105-X12).**
   - After steaming, rinse out the thickener and unfixed dye gently, at 40 °C or below with no agitation, to avoid felting.
   - Over-applied paint gives poor wet rubbing.
   - Milling acid dyes and 1:2 metal-complex dyes (Lanaset type) have better wet and light fastness than levelling acid dyes.

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2. Wool gabardine / wool twill fabric

2a. Pakistan (asked for first: nothing confirmed)

Name	Place	What	MOQ / price / lead time	Website	Fit	Status
Bannu Woollen Mills Ltd	Bannu, KP	Pakistan's oldest woollen mill (PIDC 1953). Makes tweed, blazer cloth, dress cloth, shirting, blankets. This is woollen (carded), not worsted gabardine.	None published	bwm.com.pk	Low to medium. Ask whether they can weave a worsted twill or source one.	Secondary (company and PSX filings exist, 2026 annual report)
Dawood Lawrencepur (Lawrencepur Woollen & Textile Mills)	Attock area	Historic worsted spinning and weaving unit. Operations were suspended in 2013 and the brand continued under licence.	—	—	Probably unavailable	Secondary (2013 news)
Khawaja Woolen Mills	Gujranwala	Worsted **yarn** spinner (pure wool and blends)	—	—	Possible yarn source if a Pakistani weaver makes a custom gabardine. Not realistic at 50–300 pieces.	Secondary (directory)
Fibre2Fashion "Pakistan wool fabric suppliers"	Karachi, Lahore, Rawalpindi, Faisalabad	Several wool and wool-blend listings. One is a twill from Chinese wool warp and local wool weft; one is 70/30 wool/poly at 140–350 gsm.	Price on request, login-gated	fibre2fashion.com/fabrics/pakistan-wool-fabric-suppliers-p1845-c126	Unknown. Polyester blends will not paint evenly.	Unverified
Azam Cloth Market	Lahore	Wholesale cloth hub. No source confirmed imported wool gabardine wholesale there.	One wool shirting vendor lists a minimum of 7,890 pieces	—	Low	Secondary

Practical view (my inference, unverified): Pakistani suiting brands largely import worsted cloth. The realistic route is to buy the gabardine in Italy, the EU, Turkey or China and ship it to Sialkot. Check Pakistan's import duty on wool fabric (HS 5112) against EU/UK origin rules for the finished garment.

2b. Italy / EU, including stock and deadstock platforms

Name	Place	What	MOQ / price / lead time	Website	Fit	Status
Nona Source	Paris (warehouse in France)	Deadstock from luxury maisons. B2B only (registered businesses). Wool gabardines incl. splittable, lustre RWS, stretch, fluid, Japanese 100% wool. "Ready to Dye" wool items exist (see next three rows).	Wool gabardines about €12–19/m excl. VAT. "No minimum", but no cuts: the minimum is the smallest roll listed. Professionals email their VAT number for exemption.	nona-source.com	**High for the 50-piece run** and for sampling. Stock is not repeatable, so buy the whole lot.	Verified (snippet)
Nona Source T25M13712 "Ready to Dye Virgin Wool Suiting RWS"	as above	PFD-type virgin wool suiting, dry compact hand	Weight and price not seen	nona-source.com/products/t25m13712	Possible PFD base for the overshirt. Check weight and twill.	Verified (snippet)
Nona Source T25M14298 "Ready To Dye Double Face Wool & Silk Cady"	as above	77% wool / 23% silk, 290 g/m², 134 cm	Not seen	…/t25m14298	Right weight for the trousers, but cady, not gabardine	Verified (snippet)
Nona Source T26C15006 "Ready to Dye Wool & Silk Twill Suiting RWS"	as above	86/14 wool/silk, 131 g/m², Italy	Not seen	…/t26c15006	Too light	Verified (snippet)
Zerow	Italy	Italian stock and surplus fabric platform. 100% virgin wool gabardine, 150 cm, "271 g/m". Also 99/1 wool/elastane stretch gabardine (139.6 m in 7 rolls when seen). Its only "ready-to-dye" gabardine is stretch cotton, not wool.	Wool gabardine €11.7/m (wholesale €9.4/m) excl. VAT. Sold in full rolls; minimum is the smallest roll.	zerow.it	**High for stone trousers** if 271 is g/m² (if per linear metre it is too light).	Verified (snippet)
Recovo	Barcelona	B2B deadstock marketplace, EU shipping, vendors mostly Spain and Portugal. Wool lots exist; gabardine not confirmed.	Per lot	recovo.co	Medium	Secondary
Fabricsight	EU	Deadstock from luxury maisons and ateliers	Per lot	fabricsight.com	Medium	Secondary
Vitale Barberis Canonico (VBC)	Biella	Super 110s wool gabardine (seen via J. Hilburn). Mostly men's suiting.	No stock-service terms or MOQ found. Retail resellers (e.g. Joel & Son) sell by the metre.	vitalebarberiscanonico.com	Medium. Ask the women's/fashion desk about stock-supported gabardine and whether PFD is possible.	Unverified on terms
Marzotto Group (Marzotto Wool Manufacturing, Estethia G.B. Conte, Tessuti di Sondrio, etc.)	Valdagno / Biella	Large group positioning itself as a "full-service textile company"	No stock service or MOQ found	marzottogroup.it	Medium for 300 pieces	Unverified on terms
Reda 1865	Biella	Fine worsted, mostly menswear	Nothing found	reda1865.com	Low to medium	Unverified
Drago	Biella	About 1.7 M m/yr, S130–S210, Fintes group. Stock ranges (hopsack, flannel) seen at resellers; no gabardine seen.	Nothing found	dragolanificio.com	Low to medium	Secondary
Loro Piana	Quarona	Luxury	Retail £119.90–£398.90/m (Joel & Son)	loropiana.com	**Too expensive**	Secondary
Tollegno 1900 / Botto Giuseppe / Lanificio Subalpino	Biella	Fine wool mills	Nothing on gabardine or stock	—	Unknown	Unverified

2c. UK

Name	Place	What	MOQ / price	Website	Fit	Status
Abraham Moon & Sons	Guiseley, Yorkshire	Woollen tweeds and plains, stock-supported	**1 m per design, then 10 cm steps.** Seasonal collections are not stock-supported. Trade account required.	moons.co.uk	Low: tweeds, not gabardine. Useful only as a reference	Verified (snippet, Moon FAQ)
					for a low-MOQ model.	
Dugdale Bros	Huddersfield	Merchant. Gabardines (incl. 100% wool 14–16.5 oz, too heavy). **Undyed British wool jacketing/suiting** sold as finished undyed cloth.	Not found	dugdalebros.com	The undyed jacketing could be a paintable stone base. Check weight.	Secondary
Joel & Son Fabrics	London	Retail. Stretch wool gaberdine 89% wool / 9% polyamide / 2% elastane, 150 cm, "350 g", about £79.90/m (colours up to £129.90). Also VBC and Loro Piana by the metre.	By the metre	joelandsonfabrics.com	Sampling only. Too expensive for production.	Secondary
Holland & Sherry / Marling & Evans / Bower Roebuck	Huddersfield area	Fine English worsteds	No MOQ found	—	Unknown	Unverified
Mallalieus	Delph	MOQ 120 m appears in the British Wool **Interior** guide (600 gsm upholstery), not apparel	—	—	Not relevant	Secondary
Whaleys (Bradford)	Bradford	Known UK seller of PFD fabrics. The search did **not** confirm a PFD wool gabardine.	—	whaleys-bradford.ltd.uk (verify)	Worth one email for PFD wool twill	**Unverified** (prior knowledge)
Jerseyfashion.nl (NL)	NL	Ecru "wool gabardine" 45% wool / 55% poly, 280 g/m	Retail	—	**Poor**: the polyester will not take acid dye	Secondary

2d. Turkey

Name	Place	What	MOQ	Website	Fit	Status
Yünsa	Çerkezköy (office Istanbul)	Describes itself as Europe's largest integrated worsted producer. About 10–12.5 M m/yr, mostly 100% wool. Exhibits at Première Vision.	Not published	yunsa.com/en	Medium to high for the 300-piece run	Secondary
Altınyıldız Tekstil	Istanbul / Tekirdağ	Integrated, about 9.5 M m/yr, menswear focus. Wool/poly stretch gabardine seen as US deadstock.	Not published	altinyildiz.com.tr	Medium. Ask for 100% wool.	Secondary
Unnamed Istanbul worsted makers (Fibre2Fashion)	Istanbul	Worsted fabric	Listed MOQ 100 m and 400 m	fibre2fashion.com	Unknown	Unverified

2e. China

Name	Place	What	MOQ	Fit	Status
Jiangyin Zhenxin Woollen Textile	Jiangsu	About 3.2 M m/yr worsted, tagged "Low MOQ"	Not stated	Medium	Unverified (SourceReady listing)
Jiangyin Zheyu Wool-Spinning & Weaving	Jiangsu	Spinning plus 48 rapier looms, tagged "Low MOQ"	Not stated	Medium	Unverified
Heilan Group	Jiangsu	Major worsted fabric producer	Not stated	Medium for 300 pieces	Secondary
Taizhou Yunyang Textile	Jiangsu	50/50 wool/poly gabardine, 240 gsm	MOQ "500", unit unclear	**Poor**: half polyester	Unverified (Alibaba)
Nanshan / Jiangsu Sunshine	Shandong / Jiangsu	Large worsted groups	Nothing surfaced in the search	Unknown	Unverified

3. Permanent pleating: chemistry and pleaters

3a. Chemistry (Si-Ro-Set and its successors)

Item	What	Source / supplier	Status
SI-RO-SET (CSIRO, Geelong, 1957; Farnworth et al.)	Bisulphite or reducing agent opens wool disulphide bonds and the pleat is fixed while pressed or steamed. Garment makers apply it just before final pressing, which needs equipment and QC. About 1 M wool trousers were treated in Japan in 1984/85. A CSIRO page (2011) says it was still in use.	csiropedia.csiro.au/SiroSet	Verified (snippet) as history. **Current commercial licensors or providers were not found.**
Monoethanolamine sulphite (MEAS), CAS 15535-29-2 / 17619-86-2	Classic wool "presensitise and set" agent: pad or spray, dry, then pleat and steam-press. Patents describe the commercial product as about 50% MEA bisulphite and 20% MEA sulphite.	Parchem (US chemical distributor) lists ethanolamine sulfite. Univar Solutions France lists monoethanolamine.	Verified (snippet) that the chemical is listed. No textile-grade TDS found.
Other reducing setting agents (thioglycolate, cysteine, sodium bisulphite; ethylenediamine in research)	Used for permanent creases and pleats in woven wool	Academic and patent literature	Secondary
Woolmark technical team	Current pleated and creased merino innovations ("Sculpture Merino", "Pleated Merino"). The site says to contact your local Woolmark office.	woolmark.com	Secondary. **Ask them for current pleat-setting auxiliaries and approved finishers.**

Safety: MEAS and bisulphite release SO_2. Handle them only in a pleating or finishing house with ventilation and PPE, not in a sewing unit.

3b. Pleating service houses

Name	Place	What	MOQ / lead time	Website	Fit for wool	Status
Atelier Lognon (Chanel 19M / Lemarié)	2 Place Skanderbeg, 75019 Paris (by appointment)	Hand pleating with paper moulds and heat since 1853. Outside clients have included Vanessa Bruno, Christine Phung and Opéra de Paris.	Not published	le19m.com	High quality, likely expensive. Good for development and prototypes in Paris.	Secondary
Maison du Pli (Karen Grigorian)	Paris	Independent master pleater (trained at Plissés Garbis and Gérard Lognon)	Not published	—	Development and small runs	Secondary
Omniapiega	Carate Brianza (MB), Italy	Specialist custom pleating since 1970: machine and hand plissé, soleil, origami, also sublimation. Exhibits at Milano Unica and Première Vision.	Not published	omniapiega.it	**High**. Ask about wool and chemical set.	Secondary
Plissettatura Milady	Florence	Pleating since the 1960s on "any kind of fabric": hand, cardboard-mould and machine	Not published	—	Medium to high	Secondary
Topp Italia	Bassano Bresciano	Prints, dyes, folds, plissé	Not published	—	Unknown on wool	Secondary
Ciment Pleating	UK (London area)	UK's oldest pleater (1925). Supplies designers, costumiers and colleges.	Not found	—	Medium. Ask about wool.	Secondary
Rosamanda Pleaters	Tottenham, London	One-person workshop (Kyri), hand-pattern specialist	—	—	Small runs only	Secondary (2021 profile)
Super Pleating Co.	242 Romford Rd, London E7	Listed in a college supplier list	—	—	—	Unverified (may be closed)
DSI London / Dannells	UK	Costume and lampshade pleating (2–3 week lead time at DSI)	—	—	Low	Secondary
Özlem Şardon ve Ütü San. Tic. Ltd	Turkey (city unverified)	Pleating, pressing, finishing, sanforising	—	manuzone.com	Unknown	Unverified
Shaoxing Leila Textile	Keqiao, Shaoxing, China	Pleated fabric supplier, custom work	—	—	Mostly polyester (my assumption)	Unverified
Pleating Fashion GMTS Mill (Suzhou) / Pleatskingdom (Dongguan, fewer than 5 staff)	China	Pleated-garment OEM	—	—	Unknown on wool	Unverified
Pakistan	—	**No pleating service house found.** A Karachi garment maker was seeking a computerised pleating machine. India has per-metre pleating (Surat; Mumbai with 10 m MOQ).	—	—	—	Gap
Reference only: Tom's & Sons (NY), International Pleating (US)	USA	Standard turnaround about 1 week (Tom's)	—	—	—	Secondary

Wool caveats from the sources: wool pleats need dry cleaning and pressing to stay sharp, and synthetics hold pleats better. Very creased incoming cloth delays pleaters. Most commercial pleaters heat-set polyester. Ask each one explicitly: *"Do you set 100% wool, and with what chemical assist?"*

3c. Pleating machines (if Sialkot wants to do it in-house)

- **Changzhou HuaEn:** ZJ/HE series knife, box and fancy pleating machines. Secondary.
- **Chinese exporters:** machines up to 2,900 mm width at 80 pleats/min. Secondary.
- **For wool you still need** the chemical setting step and a steam press or setting oven.

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4. Acid dyes, thickeners, steaming and painting partners

4a. Dyes

Product	Maker / place	Type	Industrial or craft	Notes	Status
Jacquard Acid Dyes	Jacquard (Rupert, Gibbon & Spider), USA	Acid dyes for wool, silk, nylon	**Craft**	Uses citric acid or vinegar (about 1 tsp per cup when painting). Gradual heating to avoid felting. Lightfastness chart on a 1–7 scale.	Verified (snippet)
Dharma Acid Dyes	Dharma Trading, USA	Acid (levelling and milling mix)	Craft	Painted acid dye needs **45–60 min steam**. Dharma itself says Dharmaset is faster on wool.	Verified (snippet)
Dharmaset (= Lanaset repack)	Dharma Trading	1:2 metal-complex plus reactive	Craft repack of an industrial range	Direct-application sheet: citric acid pre-soak 30 min or more, **steam 30–45 min**, cool, rinse, wash. "Extremely wash fast, good to excellent light fastness, better than our acid dyes on wool."	Verified (snippet)
Lanaset / Lanasol	Ex-Huntsman Textile Effects, merged into **Archroma** in 2023	Lanaset: 1:2 metal-complex, acid and reactive, pH about 4.5. Lanasol (CE): reactive, chrome-free.	**Industrial**	Huntsman's printing card covers wool printing and steaming recipes. Kraftkolour (AU) sells small quantities. **Brand continuity under Archroma not confirmed.**	Secondary
DyStar Telon (acid, levelling to milling) / Realan (reactive, metal-free)	DyStar	Wool dyes	Industrial	Pakistan distributor not found	Secondary
Kemtex Acid Dyes, Easidye, **Dischargeable** and **Illuminating** acid dyes, Decrolin	Kemtex, UK	Acid dyes (25/50/100 g), discharge system	Small professional / craft	Relevant to the indigo problem (section 1). Technical sheet DP35.	Verified (snippet)
DT Crafts acid dyes	UK	Milling acid dyes	Craft	—	Secondary
Sardar Chemical Industries (PSX: SARC)	HQ Lahore; plant at Gadoon Amazai, KP	Acid, direct and reactive dyes. Acid dyes are marketed for wool and silk. Claims to be AZO-free.	**Industrial, Pakistan-made**	Possible local supply. Ask for a wool shade card and fastness data.	Secondary
Archroma Pakistan Ltd (PSX: ARPL)	Karachi (Korangi HQ); sales offices in Lahore and Faisalabad	Dyes and auxiliaries	Industrial	Ask whether Lanaset/Lanasol-type wool dyes and printing auxiliaries are available locally	Secondary
Dylon	Consumer	Multipurpose = direct + acid + disperse mix, less wash-fast	Consumer	**Not recommended**	Secondary (Wikipedia-level)

My recommendation (inference):

- Develop with **Dharmaset/Lanaset-type** or **milling acid** dyes. These have better wet, dry-clean and light fastness than levelling acid dyes.
- For production, ask Archroma Pakistan or Sardar to match rust and caramel in an equivalent industrial range, so the dye can be bought in Pakistan.

4b. Thickeners and print-paste aids

Item	Notes	Status
Sodium alginate (Jacquard SH = high viscosity; F = low viscosity, finer line)	Standard craft painting thickener. Do **not** use the soda-ash/urea recipe meant for fibre-reactive dyes; acid dyes need acid (citric acid, or ammonium sulphate / formic acid).	Verified (snippet)
"Manutex"	Historic trade name for alginate thickeners. Current maker not confirmed.	Unverified
Industrial wool print paste (Woolwise / AWI teaching notes)	Water + thickener 2–5% (locust bean gum or alginate) + urea 0–20% + thiodiglycol or diethylene glycol 0–3% + acid donor (ammonium sulphate, or formic acid to pH 3–5) + mild oxidant (sodium chlorate) to protect dyes during steaming. Wool is often chlorinated before printing for colour yield. Acid and metal-complex prints need longer steaming than reactives.	Secondary (woolwise.com)
Guar / synthetic thickeners	Common in industry. Most published tests are on cotton.	Secondary

4c. Steaming equipment and possible steam-fix partners

Name	Place	What	Fit	Status
Zimmer Austria MODUS-E loop steamer	Klagenfurt, AT	Industrial. Silk/wool acid profile **102–104 °C, 15–30 min**.	Too large to own; useful as a spec reference	Verified (snippet, brochure)
SETeMa B-STeaM / Portafix	EU	Compact fixation steamers for start-ups and short runs	Medium. Could sit in Sialkot.	Secondary
MS Italy steamers (Ecovapor)	Italy	Digital-print steamers	Medium	Secondary
Jingjiang Huaxia small steamer (13 kW); Chinese cylindrical "star" steamers	China	Small-batch fixation; low cost; silk hung on a star frame	Medium. Cheap in-house option for Sialkot.	Secondary
Nishat Mills processing (Lahore)	Lahore	Fabric processing incl. a "drier steamer"	Fabric scale, not garments. Unlikely for 50–300 pieces.	Secondary
Digital textile print bureaus in Lahore/Faisalabad running acid inks on silk/wool	Lahore / Faisalabad	They would have steamers	**Best local bet** for steam-fixing painted flat panels	**Unverified** (no named bureau found)
Garment-dye houses that handle wool in Sialkot/Lahore/Faisalabad	—	**None found**	—	Gap

4d. Hand-painting studios and artists in Pakistan (weak leads)

Name	Place	Notes	Status
Uzma Pervez	Islamabad (2012 exhibition)	Fine artist known for hand-painted kurtas, dupattas and fabric	Secondary (old)
Fizza Asif	Lahore	Textile and fibre artist (Saatchi Art). Wearables not confirmed.	Secondary
Fahad Hussayn studio	Lahore / Karachi	Couture with heavy craft detailing. Not a painting service.	Secondary
National College of Arts (Lahore) textile dept., Indus Valley School (Karachi), Pakistan Institute of Fashion & Design (Lahore)	—	Ask for graduate textile painters or freelancers	**Unverified** (my suggestion)

5. Trims

Item	Name	Place	What / finish	MOQ / price	Website	Fit	Status
Snaps	**Scovill** (Gripper, Mighty Snaps)	USA / global	**Nickel-free finish options.** Brass Mighty Snaps and brass Gripper in **White Bronze** (a Cu-Sn-Zn, tin-based finish) are OEKO-TEX Std 100 certified. 300+ colours.	Trade	scovill.com	**High**	Verified (snippet, Scovill brochure)
Snaps	**YKK** SNAPET / BELNAP	Global; YKK Pakistan (Karachi EPZ)	"Anti-nickel" finishes and paint-to-match. AcroPlating finish.	Trade	ykkamericas.com; ykk.com.pk	High. Order may be possible via YKK Pakistan.	Verified (snippet) for the finish. Karachi supply unverified.
Snaps	**Prym Fashion**	Germany / global	Many snaps are brass with nickel finishes. "Nickel free" is only stated on some articles. L.I.F.E.-certified range.	Prym B2B: cartons of 100–200	b2b.prym.com	Medium. Request EN 1811 / EN 12472 reports per article.	Secondary
Snaps	Waris Sharif Eng. / Amir Industries	Karachi	Local snaps and zips	Unknown	—	Unknown on nickel release	Unverified
Snaps	Chinese 5-prong / 4-part, "nickel and lead free"	China (Global Sources)	Electroplated	MOQ 3,000–5,000 pcs	globalsources.com	Low to medium	Unverified
PVD finish	—	—	**No PVD snap supplier surfaced.** Ask Scovill, YKK and Prym.	—	—	—	Gap
#5 metal zip	**YKK Pakistan**	Karachi Export Processing Zone (2006 agreement; sales office listed by YKK India)	Planned scope: zips, snaps, hooks, buttons. Current output not confirmed.	Ask	ykk.com.pk	High	Secondary (old news)
Hook-and-bar	**Prym** trouser hooks: sew-on 6–12 mm, non-sew 13 mm (tool), 30 mm	Germany	Nickel, silver or black-oxidised	Trade boxes of 100 / cartons of 200	prym.com	High. Request nickel-release data if the hook touches skin.	Secondary
Cupro lining	**Bemberg** (Asahi Kasei fibre) via Tessin	Italy	100% cupro twill, 140 cm, incl. yarn-dyed (Como)	**€20/m.** Free EU shipping over €200.	tessin.it	Medium for sampling. Ask Tessin or Asahi Kasei agents for bulk.	Secondary
Cupro lining	Beyond the Pink Door (NL)	€13.70/m	—	—	—	Sampling	Secondary
Cupro lining	Joel & Son (acetate/cupro blends), The Lining Company (cupro/elastane £15.95/m, discount from 10 m)	UK	—	—	—	Sampling	Secondary
Cupro	Pakistan local Bemberg stock	—	**Not found**	—	—	—	Gap
lining							
Interfacing	**Freudenberg (Vilene)**	Global	Woven, knit and nonwoven interlinings. **No Pakistan office or agent found.** India makes woven polyester interlining.	—	apparel.freudenberg-pm.com	High quality. Ask the Asia sales office for a Pakistan agent.	Secondary
Interfacing	Pakistan Nonwoven Interlining	Korangi Industrial Area, Karachi	Local nonwoven interlining	—	—	Medium. Test on wool, under steam and in dry cleaning.	Secondary
Thread	**Coats** Epic (poly core-spun) / Gramax (textured poly)	Global	Polyester **will not take acid dye**	—	coats.com	High. Pick the colour deliberately.	Verified (product). Pakistan office not confirmed.
Thread	**Amann**	Global (Bangladesh JV)	—	—	amann.com	High. No Pakistan agent found.	Gap
Thread	Najaf Traders (Lahore, JP Coats dealer); Union Thread, Fine Thread (Pakistan)	Lahore / Pakistan	Local thread	—	—	Medium	Secondary
Woven labels	Karachi label exporter (Fibre2Fashion)	Karachi	Woven labels	**MOQ 1,000 pcs**	—	Medium	Secondary
Woven labels	Label World	Karachi	Computerised woven labels	—	—	Medium	Secondary
Woven labels	Faisalabad label makers	Faisalabad	—	MOQ about 2,000 pcs	—	Medium	Secondary
Woven labels	Patchwork Designers	Lahore	—	—	—	—	Unverified (old)
Woven labels	Superlabelstore (from 5 pcs), Dutch Label Shop, Griffes Vivienne (FR)	EU / online	Damask labels	Small MOQ	—	**High for 50-piece runs**	Secondary

Nickel-free paperwork to request:

- an EN 1811 nickel-release test;
- EN 12472 (simulated wear and corrosion) for coated parts;
- a REACH Annex XVII entry 27 declaration;
- an OEKO-TEX Std 100 certificate for the exact article and finish code.

6. Testing labs in Pakistan (ISO 105-X12 / D01 / B02)

Lab numbers are PNAC (Pakistan National Accreditation Council) accreditation numbers.

Lab	Place	Scope evidence	Status
SGS Pakistan (LAB 007)	Plot 04, Sector 24, Korangi Industrial Area, Karachi	ISO/IEC 17025. Colour fastness to rubbing, perspiration, water, sea water and **dry-cleaning** (ISO 105 / AATCC). Passed global textile proficiency testing in 2023.	Verified (snippet, PNAC). B02 not seen in the snippet.
SGS Pakistan (LAB 016)	Plot 7, Din Muhammad Town, Multan Road, Chuhang, Lahore	Accredited. Textile scope not seen.	Partial
Bureau Veritas CPS Pakistan (LAB 030)	191-S Quaid-e-Azam Industrial Estate, Kot Lakhpat, Lahore	Textile scope incl. **ISO 105-B02:2014** and **ISO 105-X12:2016**, plus pilling, abrasion and formaldehyde. Scope dated 2020.	Verified (snippet, PNAC). Check the current scope for D01.
Tti Testing Laboratories (Textile Testing International) (LAB 022), Eurofins partner	Main lab Lahore; liaison offices Karachi, Faisalabad, **Sialkot**	**ISO 105-D01:2010** (and AATCC 132) in scope (rev. dated 04-12-2024). Wider colour-fastness scope.	Verified (snippet, PNAC). Sialkot office is a liaison and sample drop point; testing is in Lahore.
Intertek Pakistan	Baig Tower, Plot E-6, Block 7&8, CCA, Karachi (+92 21 34320802-05)	PNAC "Textile Testing" listing; certificate 2024; listing date 2026	Secondary. Detailed scope not seen.
First Testing Laboratory (FTL) (LAB 210)	Harrar, Wazirabad Road, **Sialkot**	Rev. 09 (17-03-2026). Taber abrasion visible; one snippet mentions ISO 105-X12.	**Conflicting.** Check the full scope.
LAB 069 / LAB 322	Faisalabad	ISO 105-X12:2016 and B02:2014	Verified (snippet). Names not captured.
Swisstex Chemicals Technical Centre (LAB 055)	Karachi	ISO 105-X12 and B02	Verified (snippet)
Pakistan Textile Testing Foundation (PTTF)	Faisalabad	Accredited textile lab	Secondary

Suggested test pack per colourway, on the painted and steamed fabric:

- ISO 105-X12, dry and wet. Target ≥4 dry and ≥3 wet, on the painted area and the base.
- ISO 105-D01 (perchloroethylene). Target ≥4 change and staining.
- ISO 105-B02. Target ≥4 (blue wool scale) for rust and caramel.

Add these as well:

- ISO 105-E04 perspiration;
- ISO 105-E07 spotting with water. Painted acid dye can water-mark.
- A pleat-retention check after one D01 cycle.

Targets are typical brand norms, my suggestion; set your own.

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7. Briefing checklists

7a. Brief to the mill or stock platform

- [ ] 100% wool (or wool plus a small amount of elastane or polyamide only). **No polyester.**
- [ ] Gabardine or steep twill; state the twill direction you want.
- [ ] Weight: overshirt 220–240 **g/m²**, trousers 280–320 **g/m²**. Ask for usable width and g per linear metre.
- [ ] Base: PFD / ready-to-dye / ecru, **or** piece-dyed shade "indigo" and "stone" with lab-dip approval.
- [ ] If painting on indigo: can the base be dyed with **dischargeable** acid dyes? Ask for the dye class.
- [ ] Finishing declaration: no water/stain repellent, no silicone, no resin. State the anti-felt treatment if any. Is it decatised, and to what grade (KD or semi)?
- [ ] Shrinkage to steam (e.g. IWS/TM or a steam test). Wool shrinks during 30–60 min steam fixation.
- [ ] Fastness of the **base** colour to D01, X12, B02 and to steaming. The base must not bleed into the painted areas.
- [ ] MOQ per colour, stock or repeatability, lead time, sample yardage (3–5 m for painting and pleating trials), price per metre at 150 m and 800 m.
- [ ] Certifications: RWS, OEKO-TEX, origin documents for export.

7b. Brief to the pleater

- [ ] Fibre: 100% wool gabardine at the stated weights. Send the actual cloth for strike-offs.
- [ ] Pleat type: knife pleats. Give depth (e.g. 15–25 mm), spacing, direction, and the length of the pleated zone (overshirt lower half; trouser front pleats).
- [ ] Format: cut panels or fabric length? Pleat before or after painting? (Recommended: paint, steam and wash first, then pleat.)
- [ ] Setting method: paper moulds or machine, steam conditions, **chemical setting agent** (MEAS, bisulphite or other), and how the residue is neutralised and rinsed (it causes odour).
- [ ] Any effect of setting chemistry or steam on painted colours. Ask for a test on painted swatches.
- [ ] Durability: pleat retention after dry cleaning (D01) and steam pressing, and after a damp-cloth test.
- [ ] Width take-up factor, so fabric can be ordered correctly (knife pleats typically take about 3×).
- [ ] MOQ, price per panel or per metre, lead time, transit packaging that keeps pleats.

7c. Brief to the dye, paint or steam house

- [ ] Dye class: milling acid or 1:2 metal-complex (Lanaset type), or Kemtex dischargeable/illuminating for indigo. Give shade standards for rust and caramel.
- [ ] Paste recipe: thickener (alginate, locust bean gum or guar) at a stated %, urea, thiodiglycol or DEG, acid donor (ammonium sulphate or formic acid, pH 3–5), optional sodium chlorate.
- [ ] Application: brush, sponge or stencil. Set the maximum paste load per m² (over-loading ruins wet rubbing).
- [ ] Drying, then **saturated steam 100–104 °C for 30–60 min** (trial times). Protect with cotton or newsprint interleaves so colour does not transfer.
- [ ] Wash-off: cold rinse, then ≤40 °C rinse with a wool-safe detergent, no agitation, flat dry. Check felting and shrinkage.
- [ ] Order relative to fusing, pleating and sewing. Run steam tests on fused panels.
- [ ] Lab tests: X12, D01, B02, E04, E07 on the steamed and washed swatches **before** bulk.
- [ ] Capacity per day, cost per piece, rejects policy, photo standard for "hand-painted variation".

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8. Biggest gaps and suggested next steps

1. **No Pakistani worsted gabardine source was confirmed.** Plan to import:
   - Nona Source or Zerow for the 50-piece run;
   - Yünsa or Italian mills for the 300-piece run.
2. **No Pakistani pleater or wool steam-fix house was confirmed.** Two options:
   - pleat in Europe (Omniapiega, a Paris pleater, or a UK pleater) and ship panels to Sialkot; or
   - set up a small steamer plus MEAS pleat-setting in Sialkot with Woolmark or Archroma technical support.
3. **Indigo plus rust/caramel acid paint is a colour-physics problem,** not a sourcing one. Decide between a tonal effect and the discharge route before buying cloth.
4. **Labs:** Tti (Lahore, with a Sialkot liaison office) covers D01. BV Lahore covers B02 and X12. SGS Karachi covers dry-cleaning and rubbing. Confirm current scopes on PNAC.

Sources

(All accessed via web search, 11 Nov 2026. Direct page fetches were blocked in this environment.)

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